

# Cybersecure and Cyber-Resilient European Cloud–Edge–IoT Computing Continuum: A Strategic Roadmap

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**ABSTRACT** The distribution of computation across cloud, edge, and IoT tiers, combined with the simultaneous operationalisation of NIS2, the Cyber Resilience Act, the AI Act, and DORA, has made cybersecurity, resilience, and digital sovereignty an integrated strategic problem for Europe. Perimeter-oriented security models do not scale to heterogeneous, physically exposed, and multi-jurisdictional continuum infrastructures, and Europe's normative leadership in trust, privacy, and certification translates into strategic advantage only when embedded in deployable capabilities. We term this distance the ambition–implementation gap. This paper addresses that gap through a three-stage methodology for identifying and prioritising cybersecurity capabilities: architectural decomposition of the continuum into six Security Layers and four Transversal Security Cross-Layers together with their constituent capabilities (Stage 1); consolidation of those capabilities into three executive priority clusters, namely Sovereign Trusted Infrastructure, Cognitive Security and Autonomous Cyber Defence, and Federated Operational Sovereignty, analysed through a seven-dimension synthesis template (Stage 2); and projection of each cluster onto a 2026–2040 transition matrix anchored by Operational Deployment Phase and Strategic Sovereignty Level bands and terminating in a single Critical Enabler per phase column (Stage 3). The methodology integrates a Sovereignty Level construct as an interpretive strategic lens. From the cluster trajectories and a phase-gate readiness assessment, the paper derives seven implementation-oriented recommendations (three cluster-aligned and four cross-cutting), each tracing quantified Phase 2 and Phase 3 targets to the gate condition on which delivery depends. The contribution is methodological and strategic: it provides a reproducible template for cybersecurity roadmap construction under ecosystem uncertainty and frames autonomous, federated, and certified Security-by-Design as the operational target for European digital infrastructure.

**INDEX TERMS** Certification, Cloud–Edge–IoT continuum, confidential computing, Critical Enabler, cyber-resilience, digital sovereignty, European Union, federated infrastructure, post-quantum cryptography, security roadmap, supply-chain attestation, Zero-Trust architecture.

## I. INTRODUCTION

The shift from centralised cloud to a distributed Cloud–Edge–IoT continuum relocates computation, storage, and inference closer to devices, users, and physical processes [1]–[3]. These architectural gains, including lower latency, higher operational autonomy, and locality of data processing, come with a structural cost. The attack surface expands, system heterogeneity increases, and defenders must reason across a larger number of administrative and trust domains. Perimeter-oriented security models designed for consoli-

dated data-centre environments are therefore inadequate for infrastructures that are geographically dispersed, resource-constrained, and physically exposed [4]–[6]. Each relocation of workload across providers and jurisdictions is also a transfer of control, which makes continuum security inseparable from digital sovereignty, understood here as the ability to govern, audit, and operate critical infrastructure under credible European influence. This coupling is sharpened by three concurrent global trends: structural hyperscaler concentration in the EU cloud market [7], migration to post-quantum

cryptography under a now-binding European framework [8], and the weaponisation of generative AI against both human operators and AI-enabled defences.

Europe faces this transition under four simultaneous regulatory activations. NIS2 [9], the Cyber Resilience Act (CRA) [10], the AI Act [11], and DORA [12] together raise the baseline of mandatory security, resilience, accountability, and supply-chain obligations across essential entities, ICT products, high-risk AI systems, and financial-sector third-party providers. The EU cybersecurity certification framework [13], [14] provides the assurance substrate that these regulations presuppose. Operationalisation of this combined regime is uneven: national transposition lags the legal timetable in several large Member States [15], EUCS remains unadopted [16], and conformity-assessment infrastructure for AI-specific security obligations is not yet in place [17]. Section II provides detailed evidence on these patterns and on the asymmetry between Europe’s regulatory leadership and its industrial position.

The central argument of this paper is that Europe’s normative leadership in trust, privacy, governance, and certification [18], [19] translates into strategic advantage only when it is embedded in deployable capabilities. We call the distance between normative leadership and deployable capacity the ambition–implementation gap. Closing this gap requires a transition from reactive and siloed security towards autonomous, federated, and certified Security-by-Design: infrastructures that sustain trustworthy operation under adverse conditions, security mechanisms that span providers and jurisdictions, and assurance embedded throughout the system lifecycle [4], [20], [21].

In this paper, a roadmap is understood as a structured strategic instrument that links current capability gaps to future target states through phased actions, dependencies, and transition conditions. It is not a forecast of technological development, but a coordination tool for aligning research, innovation, policy, standardisation, deployment, and ecosystem investment over time [3]. The Cloud–Edge–IoT continuum refers to the distributed execution environment in which computation, storage, networking, sensing, and AI inference are dynamically placed across central cloud platforms, intermediate edge infrastructures, and resource-constrained devices [1]–[3]. The term continuum is used because these tiers are increasingly interdependent rather than isolated: workloads, data flows, trust decisions, and control functions move across them according to latency, resilience, privacy, energy, and sovereignty requirements [1], [22]. The term ecosystem captures the fact that secure operation depends not only on technologies, but also on providers, standards bodies, regulators, certification schemes, open-source communities, research infrastructures, SMEs, large industrial actors, and public authorities whose decisions jointly determine deployment capacity [3], [13], [14].

The paper makes five contributions. First, it introduces a three-stage methodology for constructing cybersecurity roadmaps under conditions of ecosystem uncertainty, com-

bining architectural decomposition (Stage 1), priority clustering with seven-dimension analysis (Stage 2), and phased transition-matrix projection (Stage 3), while integrating a Sovereignty Level construct as an interpretive strategic lens (Section III). Second, it provides an architectural decomposition of the Cloud–Edge–IoT continuum into six Security Layers and four Transversal Security Cross-Layers, populated with their constituent capabilities, as the Stage 1 output (Section IV). Third, it consolidates those capabilities into three priority clusters: Sovereign Trusted Infrastructure, Cognitive Security and Autonomous Cyber Defence, and Federated Operational Sovereignty. Each cluster is analysed through a seven-dimension template as the Stage 2 output (Section V). Fourth, it develops a three-phase transition matrix for each priority cluster, projecting the seven-dimension analysis onto the 2026–2040 horizon and identifying, for each phase boundary, the Critical Enabler whose delivery determines whether progression to the next phase is plausible, as the Stage 3 output (Section VI). Fifth, it derives a set of seven implementation-oriented recommendations, including three cluster-aligned and four cross-cutting recommendations, from the cluster trajectories and phase-gate readiness assessment, with quantified Phase 2 and Phase 3 targets traced to the gate conditions on which delivery depends (Section VII).

Section II provides the strategic and regulatory context. Section III presents the three-stage methodology. Section IV defines the layered system model and its constituent capabilities. Section V presents the three priority clusters and their seven-dimension synthesis. Section VI presents the phase logic and cluster transition matrices. Section VII states the recommendations. Section VIII discusses validation and limitations. Section IX concludes.

## II. BACKGROUND AND DRIVING FACTORS

The first vector shaping the continuum transition is architectural: the continuum is characterised by decentralisation and physical exposure. Decentralisation complicates visibility, orchestration, patching, identity, and incident response across multiple administrative domains and increases the probability of inconsistent security postures across deployments [23], [24]. Physical exposure removes the environmental and access controls that hyperscale data centres rely on: edge nodes in factories, substations, transport hubs, and telecom sites are exposed to tampering, fault injection, side-channel attacks, and supply-chain compromise [6]. Hardware roots of trust, secure boot, remote attestation, and tamper-aware design are therefore baseline requirements rather than optional enhancements [25], [26].

The European regulatory landscape constitutes a second major vector shaping the continuum transition. NIS2 [9], the CRA [10], the AI Act [11], DORA [12], the Data Act [27], and GDPR [28] together impose obligations that must be translated into architectures, assurance mechanisms, and lifecycle controls operating across cloud, edge, and IoT. This simultaneity creates disproportionate implementation

pressures for SMEs and smaller Member States.

Sovereignty is the third vector. Dependence on non-EU providers for secure silicon, cloud platforms, and enterprise security tooling creates geopolitical and operational risk. Extra-territorial data access under foreign jurisdiction has been confirmed as a practical constraint for EU users of hyperscaler services [29].

Europe's position in this landscape is asymmetric. A world-leading regulatory foundation and a rigorous certification framework [13], [14] coexist with structural industrial gaps. EU-headquartered vendors (Infineon, NXP, STMicroelectronics) design the dominant secure-element products and achieved EUCC "High" certification for multiple product families in 2025 [30], but fabrication depends on non-EU foundries and designs remain proprietary, preventing sovereign auditability at the silicon level. Intel SGX/TDX, AMD SEV-SNP, and Arm TrustZone/CCA constitute the dominant operational hardware-backed TEE families across cloud, server, and embedded deployments; EU-origin RISC-V alternatives remain largely at research or prototype stage [31], and no EU-origin RISC-V processor has received EUCC or Common Criteria certification as of April 2026. The enterprise SIEM market is dominated by US platforms, with no EU-headquartered vendor in the Gartner Leaders quadrant as of late 2025 [32]. Hyperscalers hold approximately 70% of the EU cloud market; EU-headquartered providers collectively hold about 15%, stable since 2022 but below the 29% share recorded in 2017 [7]. For post-quantum migration the NIS Cooperation Group roadmap [8] now provides a binding framework, but production-scale post-quantum cryptographic (PQC) hardware tailored for edge and IoT is not yet available in Europe. This asymmetry between regulatory and certification leadership on the one hand, and structural industrial gaps on the other, is the empirical instantiation of the ambition–implementation gap that Section I introduces and that the methodology of Section III is designed to address.

### III. METHODOLOGY

The methodology examines the heterogeneous landscape of the Cloud–Edge–IoT ecosystem and translates it into a structured roadmap for research, innovation, policy, deployment, and ecosystem development. It proceeds in three stages, followed by iterative review and refinement. Stage 1 (Section III-A) decomposes the continuum into architectural layers and cross-layers and identifies the specific capabilities required within each. Stage 2 (Section III-B) consolidates those capabilities into priority clusters and analyses each cluster through a seven-dimension template. Stage 3 (Section III-C) projects the priority clusters onto three-phase transition matrices covering the 2026–2040 horizon, identifying the Critical Enabler associated with each phase boundary.

This separation is deliberate. Stage 1 is a decompositional operation: it establishes the analytical vocabulary, exposes cross-layer dependencies, and prevents priorities from being selected as isolated technologies detached from their

architectural context. Stage 2 is a synthetic operation: it aggregates capabilities that share dependencies or strategic outcomes into coherent investment and coordination domains and characterises each through a static seven-dimension profile. Stage 3 is a projective operation: it embeds those static profiles in time, exposing the sequencing of research, regulation, deployment, and market uptake required to move each cluster from its 2026 baseline to its 2040 target. Critical Enablers, phase-gate conditions, and shared bottlenecks are products of Stage 3 rather than of Stage 2, because they only become visible once the cluster is examined as a temporal trajectory.

Rather than predicting when each technology or policy milestone will occur, the methodology organises the capabilities, dependencies, and transition conditions that make progress plausible. This distinction is important because the Cloud–Edge–IoT ecosystem evolves through the interaction of technical maturity, regulatory obligations, deployment readiness, market adoption, and industrial capacity. The methodology also links technical cybersecurity capabilities to broader European strategic objectives: resilience, interoperability, trust, and sovereignty.

In this paper, Sovereignty Level denotes the extent to which critical capabilities and dependencies can be shaped, governed, assured, and operated under European influence or control. It is assessed at the capability and dependency level in Stage 1, used to interpret the strategic significance of priority clusters in Stage 2, and projected as a per-phase qualitative band in Stage 3 (the SSL row of each transition matrix). It is used as an interpretive strategic lens rather than as a rigid quantitative score.

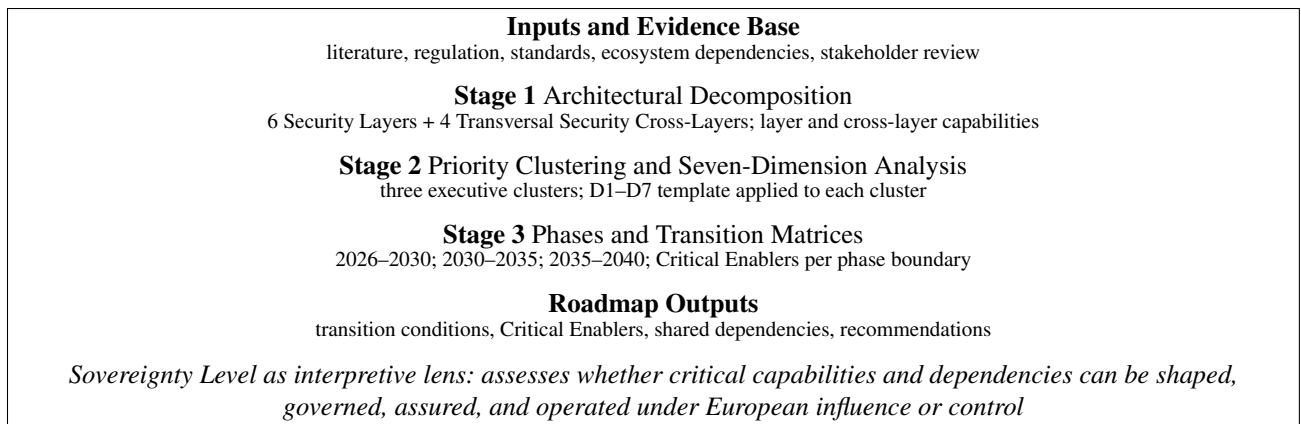
#### A. STAGE 1: ARCHITECTURAL DECOMPOSITION AND CAPABILITY IDENTIFICATION

Stage 1 decomposes the continuum into architectural vertical and cross-cutting layers. These layers are treated as analytically distinct but operationally interdependent: a vulnerability or governance gap at one layer typically propagates to adjacent layers, so effective security requires coordinated action across the vertical and horizontal extent of the architecture [6], [24]. For each layer, Stage 1 further identifies the constituent capabilities: the technical and operational subdomains in which strategic investment, standardisation, and ecosystem building are required.

Stage 1 answers three questions:

- 1) Where are the major security and resilience problem domains within the continuum?
- 2) At which architectural layers do they emerge?
- 3) Which capabilities within each layer are sufficiently strategic, cross-cutting, or underdeveloped to justify dedicated roadmap attention?

Capabilities are derived from four sources: the architectural characteristics of distributed systems across HPC, cloud, edge, and device tiers; the recurring cybersecurity challenges observed at those tiers (hardware integrity, software supply-chain verification, workload isolation, identity



**Figure 1.** Three-stage methodology for constructing the cybersecurity and cyber-resilience roadmap. Stage 1 decomposes the Cloud–Edge–IoT continuum into Security Layers, Transversal Security Cross-Layers, and constituent capabilities. Stage 2 consolidates these capabilities into priority clusters and analyses each through a seven-dimension template. Stage 3 projects the clusters onto three-phase transition matrices, each row reflecting one of four assessment perspectives and the Critical Enabler row identifying the gate condition for phase progression. Sovereignty Level is used as an interpretive lens across capability, dependency, cluster, and phase assessment.

federation, data confidentiality, and perimeter control); cross-cutting strategic concerns (sovereignty, trust, interoperability, resilience, assurance, and regulatory alignment); and the need to connect technical architecture with operational constraints, governance requirements, and deployment pathways. A capability is therefore defined as a strategic ability that the European ecosystem should be able to design, deploy, govern, standardise, assure, coordinate, or scale in order to achieve trustworthy, resilient, and sovereign operation of the continuum. Capabilities may encompass technical controls, governance mechanisms, interoperability instruments, assurance functions, ecosystem coordination capacities, and deployment enablers.

The output of Stage 1 is a structured map of the problem space, organised into six Security Layers and four Transversal Security Cross-Layers, each populated with its constituent capabilities (see Section IV). Sovereignty-sensitive capabilities, where dependencies on external providers or assurance mechanisms are structurally significant, are identified so that Stage 2 can track their evolution.

## B. STAGE 2: PRIORITY CLUSTERING AND SEVEN-DIMENSION ANALYSIS

Stage 2 groups the capabilities identified in Stage 1 into three Priority Clusters: (i) Sovereign Trusted Infrastructure, the hardware, cryptographic, software-update, data-protection, and resilience foundations that anchor trust across the continuum; (ii) Cognitive Security and Autonomous Cyber Defence, the AI-assisted operational layer that scales detection, analysis, and response beyond human-paced workflows; and (iii) Federated Operational Sovereignty, the interoperability, supply-chain verification, assurance, and federation substrate that converts individually sovereign components into a unified European trust fabric.

The use of three priority clusters is a deliberate synthesis choice rather than an assertion that only three cybersecurity priorities exist. This is consistent with technology-

roadmapping practice, where roadmaps function as strategic planning and communication instruments that organise complex technical trajectories into a smaller number of decision-relevant domains [33]. A roadmap may identify many relevant capabilities, but policy and funding coordination require higher-level priorities that communicate strategic direction, expose shared dependencies, and support portfolio-level decision-making. The three clusters therefore function as an executive structuring layer: they consolidate related capabilities into coherent investment and coordination domains while preserving the more granular capability analysis underneath.

Each priority cluster is analysed through a seven-dimension template applied consistently across the three clusters. The same template also preserves traceability to the underlying capabilities from which each cluster is derived:

- **(D1) Baseline**, describing the current state of the priority cluster, its capability maturity, dependencies, and main implementation gaps;
- **(D2) High-Impact Activities and Target Outcomes**, identifying the concrete research, innovation, standardisation, or deployment actions expected to advance the priority cluster, with each action assigned to a target phase and linked to the intended target outcome;
- **(D3) Primary Drivers**, identifying the threat and regulatory drivers that make the priority cluster urgent and non-deferrable;
- **(D4) Catalysts and Enablers**, identifying the funding instruments, regulatory and standardisation mechanisms, pilot infrastructures, and ecosystem coordination mechanisms that can accelerate the priority cluster;
- **(D5) Disruptors and Systemic Risks**, identifying technical cliffs such as the emergence of cryptographically relevant quantum computers, dependency risks such as supply-chain exposure and vendor lock-in, market concentration, and geopolitical constraints that could delay, weaken, or redirect the priority cluster;



- **(D6) Failsafe Mechanisms**, specifying fallback, isolation, rollback, redundancy, and graceful-degradation mechanisms that preserve security and resilience if the primary capability or deployment pathway is delayed, compromised, or only partially realised; and
- **(D7) Transversal Meta-Layer**, assessing how the priority cluster contributes to broader European strategic objectives, including sovereignty, trust and ethics, and sustainable security.

D2 lists the activities that should advance a cluster but treats them as a static portfolio. D4 identifies the broader funding, standardisation, and pilot infrastructures that can accelerate the cluster as a whole. The temporal sequencing of these activities and the identification of the specific gate conditions whose delivery determines phase progression are operations of Stage 3, not Stage 2. The seven-dimension template therefore provides a structured and sufficiently comprehensive static profile of each priority cluster, enabling cross-cluster comparison, funding prioritisation, and the identification of shared dependencies, while leaving the temporal projection to the next stage.

### C. STAGE 3: PHASE LOGIC AND TRANSITION MATRICES

Stage 3 projects each priority cluster onto a three-phase transition matrix covering the 2026–2040 horizon. The phases are:

- **Phase 1: Near-Term Implementation and Pilot Actions (2026–2030)**, focused on deploying near-ready solutions, enforcing enacted regulation, validating early pilots, and closing immediate governance gaps;
- **Phase 2: Mid-Term Scale-Up, Industrial R&D and Standardisation (2030–2035)**, focused on industrialising sovereign capabilities, federating architectures, scaling validated pilots, and building the workforce and institutional infrastructure required for wider adoption; and
- **Phase 3: Long-Term Strategic Autonomy and Leadership (2035–2040)**, focused on achieving operational maturity, consolidating European leadership, advancing frontier research, and resolving long-term governance and oversight questions.

#### 1) Assessment Perspectives

The content of each phase column is informed by four assessment perspectives. The four are chosen because they jointly capture the distinct ecosystem conditions that must align for a cluster to progress, while being narrow enough to remain analytically tractable. Technical readiness alone does not predict adoption; regulatory obligation alone does not produce deployment; deployment alone does not produce market viability. The four perspectives are therefore treated as concurrent rather than sequential:

- **R&I Maturity**, covering the technical maturity of the capabilities underlying the priority cluster, the maturity

of their enabling dependencies, the availability of pilot deployments, and the readiness of the research and innovation ecosystem to support transition;

- **Policy, Regulation, and Standards**, covering the regulatory obligations that drive the priority cluster, the standards and certification mechanisms that support it, and the policy instruments that can accelerate or constrain adoption;
- **Operational Deployment and Adoption**, covering the availability of deployable solutions, the maturity of deployment pathways, and the readiness of users, operators, and infrastructure providers to adopt the priority cluster; and
- **Ecosystem Uptake and Strategic Impact**, covering the ecosystem conditions that shape adoption, the strategic significance of achieving the target state, and the consequences of delay or failure.

These four perspectives are not represented as separate rows in the compact transition matrices presented in Section VI, but are folded into the Operational State row of each matrix, which characterises the observable state of the cluster within each phase column across all four perspectives concurrently. The narrative cluster analyses in Section V expand each perspective in prose.

Throughout the paper, phase gate refers to the checkpoint between two consecutive phases, the Critical Enabler names the single condition whose delivery controls that gate, and gate condition is used as a generic descriptor where the abstraction is intended rather than a specific instance.

#### 2) Operational Deployment Phase and Strategic Sovereignty Level Rows

Each transition matrix uses two scalar rows to position the cluster within each phase. The Operational Deployment Phase (ODP) row uses the fixed vocabulary *Applied R&D* → *Fragmented Piloting* → *Standardised Integration* → *Ubiquitous Operation*, and characterises how broadly and how operationally the cluster is deployed. The Strategic Sovereignty Level (SSL) row uses the fixed vocabulary *High Foreign Dependency* → *Emerging EU Ecosystem* → *Full Sovereign Autonomy*. The SSL row operationalises, at the cluster level and per phase, the same Sovereignty Level construct introduced as an interpretive lens in Section III: it is the per-phase expression of that lens, not an independent metric. Where a cluster contains capabilities at heterogeneous sovereignty positions (Cluster 3 in Phase 1 being the principal example), the SSL cell records that heterogeneity rather than smoothing it into a single band.

#### 3) Critical Enabler Row

Each phase column terminates in a Critical Enabler cell. The Critical Enabler is a single named gate condition whose delivery is treated as necessary for the cluster to progress to the next phase. It is distinguished from the broader Catalysts and Enablers of Stage 2 (D4) in two respects. First, where D4 lists the funding instruments, standards bodies,

| Execution environments                      |       |      |         |
|---------------------------------------------|-------|------|---------|
| HPC                                         | Cloud | Edge | Devices |
| L6: Network and Perimeter                   |       |      |         |
| L5: Data-Centric (PQC, TEE, QKD, SMPC)      |       |      |         |
| L4: User and Orchestration (ZT, IAM, PAM)   |       |      |         |
| L3: Application (Workloads, APIs)           |       |      |         |
| L2: Software (OS, Middleware, OTA)          |       |      |         |
| L1: Hardware (RoT, PhySec)                  |       |      |         |
| CL1: Cyber-Resilient SW/AI Lifecycle        |       |      |         |
| CL2: Dynamic Risk and SecOps                |       |      |         |
| CL3: Interoperability (Simpl, CAMARA, MIMs) |       |      |         |
| CL4: Compliance-by-Design and Assurance     |       |      |         |

**Figure 2.** Security framework: four execution environments, six Security Layers (vertical stack, L1–L6), and four Transversal Security Cross-Layers (CL1–CL4).

and pilot infrastructures that can accelerate a cluster as a whole, the Critical Enabler names the single condition whose absence would block phase transition irrespective of the wider catalyst landscape. Second, where D4 entries are static contextual enablers, the Critical Enabler is an event with a phase-bound delivery expectation: a regulatory enforcement date, a standard finalisation, a fabrication-line milestone, or an equivalent named occurrence. The Critical Enabler is therefore the principal phase-gate instrument of the roadmap, and the assessment of whether it is on track, at risk, or blocked is the principal analytical output of Stage 3.

A transition condition expressed through the Critical Enabler row indicates that a technical basis has matured, that a policy or standardisation basis has emerged, that a deployment pathway has been validated, that a strategic ecosystem condition has materialised, or that a Sovereignty Level improvement has become plausible through reduced dependency or increased European capacity. These conditions are not deterministic predictions or precise quantitative thresholds unless the available evidence supports such specificity.

#### IV. STAGE 1: CLOUD-EDGE-IOT CORE DIMENSIONS (LAYERS) AND CAPABILITIES IDENTIFICATION

The continuum is modelled as a layered reference architecture spanning four execution environments (HPC, Cloud, Edge, Devices) and organised into six *Security Layers* forming the vertical stack and four *Transversal Security Cross-Layers* acting as horizontal enablers, as shown in Figure 2. HPC refers to infrastructure for large-scale scientific and AI training capacity; Cloud offers elastic service hosting and orchestration; Edge brings computation closer to users, industrial sites, telecom infrastructure, and data sources; and Devices include resource-constrained sensors, actuators, embedded systems, and IoT endpoints operating closest to the physical world. The decomposition aligns with established reference architectures, including ETSI MEC GS 003 [22], NIST SP 800-207 and SP 800-207A for Zero Trust [4], [5], NIST CSF 2.0 [34], ISO/IEC 27400 [35], and the ENISA baseline for IoT [36], and is consistent with the distributed computing-continuum framing in recent literature [1], [2] and in the European industrial technology roadmap for the next-

generation cloud-edge [3].

The layers are analytically distinct but operationally inter-dependent: gaps at one layer can propagate to adjacent layers, as documented in the edge-computing security literature [6], [23], [24]. The model provides the shared vocabulary used throughout the paper and exposes cross-layer dependencies, allowing the priority clusters in Section V and the transition matrices in Section VI to be analysed as an interconnected portfolio rather than as isolated priorities. The EU cybersecurity certification framework [13] and its EUCC operationalisation [14] provide the regulatory anchoring for several of the capabilities defined below.

The remainder of this section introduces each layer and cross-layer by describing its purpose within the security framework and listing its constituent capabilities. Table 1 consolidates the capabilities, their layer or cross-layer codes, and their principal references. Each capability is treated as an atomic analytical unit; the three priority clusters analysed in Section V are obtained by aggregating and translating these capabilities into actionable strategic needs.

##### A. SECURITY LAYERS (VERTICAL STACK)

The six Security Layers define the vertical stack of the continuum, moving from physical and silicon-level trust foundations through software, applications, identity and orchestration, data protection, and the network infrastructure. They are presented from the lowest trust anchor upwards because weaknesses at lower layers can undermine guarantees provided by higher layers, while higher-layer controls are required to operationalise security, governance, and resilience across distributed environments.

###### 1) Layer 1: Hardware-Centric Security

Layer 1 addresses the physical and silicon-level trust foundations of the continuum and provides the verifiable anchors on which higher-layer security assurances depend. It covers hardware roots of trust, secure boot, remote attestation, Trusted Platform Modules, Physical Unclonable Functions, and physical-layer detection primitives [25], [26]. Without a verifiable silicon anchor, software-level attestation and cryptographic identity rest on assumptions that are difficult to validate in distributed and physically exposed environments. Layer 1 is the principal regulatory locus of hardware sovereignty in the continuum, and it is the layer at which EUCC “High” certification [14] operates.

**Capability L1.1, Hardware Roots of Trust.** Cryptographically verifiable device and supply-chain integrity from fabrication to deployment, with a strategic focus on advancing European open-hardware designs (RISC-V) toward EUCC “High” assurance. Technical foundations derive from TPM 2.0 [25] and integrity measurement architectures [26]. European instances include OpenTitan [37], Keystone [38], and the DARE SGA1 RISC-V initiative [39].

**Capability L1.2, Physical-Layer Threat Detection and Fingerprinting.** Exploitation of two complementary classes of physical-layer features: (i) channel-intrinsic observables,

**Table 1.** Layer and cross-layer capabilities with their principal references.

| Code  | Capability                                                        | Primary references     |
|-------|-------------------------------------------------------------------|------------------------|
| L1.1  | Hardware Roots of Trust; EU open-hardware (RISC-V) to EUCC “High” | [25], [26], [37]–[39]  |
| L1.2  | Physical-layer threat detection and RF fingerprinting             | [6], [24], [40]        |
| L2.1  | OS and middleware hardening; sovereign edge Linux                 | [4], [41]              |
| L2.2  | Secure OTA patch and update management                            | [42], [43]             |
| L3.1  | Workload security (service mesh, mTLS, sidecar)                   | [44], [45]             |
| L3.2  | In-band network telemetry and deep observability                  | [44], [45]             |
| L3.3  | API security gateway and policy enforcement                       | [41], [46]             |
| L4.1  | Zero Trust orchestration (PEP/PDP)                                | [4], [5]               |
| L4.2  | Federated IAM and SSI                                             | [47]–[49]              |
| L4.3  | Privileged Access Management and JIT                              | [4], [5]               |
| L5.1  | Confidential computing and TEEs                                   | [38], [50]             |
| L5.2  | Post-Quantum Cryptography for edge/IoT                            | [8], [51]–[54]         |
| L5.3  | Signalling/data-plane cryptographic authentication                | [55]–[58]              |
| L5.4  | Distributed reputation and trust networking (DLT)                 | [13], [59]–[61]        |
| L5.5  | QKD for critical links                                            | [62], [63]             |
| L5.6  | Privacy-preserving computation: SMPC and homomorphic encryption   | [11], [28], [64]–[66]  |
| L6.1  | Telco-cloud and O-RAN security                                    | [55]–[58]              |
| L6.2  | Physical resilience and IT/OT safety                              | [6], [24], [55], [58]  |
| L6.3  | Edge-native autonomous guardians                                  | [56]–[58]              |
| L6.4  | Non-terrestrial network and satellite security                    | [67], [68]             |
| CL1.1 | Continuous DevSecOps                                              | [41]                   |
| CL1.2 | Model integrity and DataBOMs                                      | [69], [70]             |
| CL1.3 | Dynamic supply-chain verification; Runtime SBOMs                  | [71]                   |
| CL1.4 | Automated adversarial testing and purple teaming                  | [72]                   |
| CL2.1 | Automated SOC and collaborative CTI                               | [73]                   |
| CL2.2 | High-precision telemetry and security signalling                  | [74], [75]             |
| CL2.3 | Internet of Behaviour and evidence sharing                        | [75]–[78]              |
| CL2.4 | AI-native analytics (NWDAF, LLMSecOps)                            | [69], [70], [74]       |
| CL2.5 | CTI lifecycle management                                          | [75]–[78]              |
| CL2.6 | Deception technology and active defence                           | [75], [76]             |
| CL3.1 | Simpl integration and heterogeneous interoperability              | [59]–[61]              |
| CL3.2 | Neutral host and multi-tenant isolation                           | [4], [5]               |
| CL3.3 | Pan-European NaaS federation                                      | [79], [80]             |
| CL3.4 | Semantic security interoperability                                | [60], [61]             |
| CL3.5 | Cyber range and digital twin federation                           | [3], [13], [73]        |
| CL4.1 | PQC readiness and Zero Trust architectures                        | [4], [5], [51]–[53]    |
| CL4.2 | Continuous automated certification (EUCC; EUCS)                   | [13], [14], [81], [82] |
| CL4.3 | Multicloud Key Management as a Service                            | [4], [5], [51]–[53]    |
| CL4.4 | Cross-Jurisdictional Federated AI Governance                      | [27], [28], [50]       |
| CL4.6 | Cyber risk quantification and board-level reporting               | [9], [12], [34]        |
| CL4.7 | AI trustworthiness and algorithmic accountability                 | [11], [69], [70]       |

including Channel State Information (CSI), Carrier Frequency Offset (CFO), and Received Signal Strength Indicator (RSSI); and (ii) transmitter-intrinsic Radio Frequency Fingerprints (RFF) arising from device-specific hardware impairments such as I/Q imbalance, power-amplifier non-linearity, and oscillator phase noise. Together these enable real-time device authentication and anomaly detection at the lowest communication layer, complementing upper-layer cryptographic protections and providing identity assurance for resource-constrained devices that cannot reliably execute conventional authentication protocols [6], [24]. Operational deployment of RFF at the European edge, however, is gated by two coupled trustworthiness challenges: feature drift under temporal, thermal, and propagation-environment variation, which degrades classifier accuracy over the device life-cycle; and the privacy exposure inherent to persistent, non-revocable hardware-level identifiers, which engages GDPR Article 25 (data protection by design and by default) and motivates privacy-preserving RFF pipelines as a precondition

for deployable physical-layer security [40].

## 2) Layer 2: Software-Centric Security

Layer 2 secures the software substrate on which continuum workloads execute, including hypervisors, container runtimes, distributed operating systems, and the pipelines that maintain them. Its central concerns are attack-surface reduction through OS and middleware hardening, and cryptographically verified, failure-resilient software distribution to field-deployed nodes for which physical access for remediation is limited or infeasible [42], [43]. Assurance at Layer 2 is necessary for trustworthy execution of the services and workloads built above it.

**Capability L2.1, OS and Middleware Hardening.** Security hardening of hypervisors (KVM/QEMU), container runtimes, and distributed operating systems, with emphasis on sovereign, lightweight Linux distributions optimised for edge resource constraints and reduced attack surface, in line with DevSecOps and Zero Trust guidance [4], [41].

**Capability L2.2, Secure OTA Patch and Update Management.** Cryptographically verified, failure-resilient over-the-air update pipelines for field-deployed edge nodes such as smart grids, O-RAN radios, and industrial gateways. The IETF SUIT architecture [42] and the Uptane standard [43] provide the main reference frameworks.

### 3) Layer 3: Application-Centric Security

Layer 3 protects the workloads and service interfaces executing across the continuum, including containerised applications, serverless functions, service meshes, and the APIs that expose them. Application-layer defences are a key enforcement point for obligations that surface as runtime behaviours rather than infrastructure properties, including AI Act [11] and GDPR Article 25 [28] requirements. The NIST SP 800-204 microservices security series [41], [44], [45] and the OWASP API Security Top 10 [46] provide the main reference controls.

**Capability L3.1, Workload Security.** Securing containerised applications and serverless functions through hardened service meshes and sidecar proxies that provide mutual TLS, traffic policy enforcement, and granular inspection while preserving operational performance, aligned with NIST SP 800-204 [44] and SP 800-204A [45].

**Capability L3.2, In-Band Network Telemetry and Deep Observability.** Embedding telemetry directly into packets or service-mesh flows traversing containerised workloads, enabling real-time security monitoring and anomaly detection without exclusive reliance on dedicated out-of-band monitoring infrastructure [44], [45].

**Capability L3.3, API Security Gateway and Policy Enforcement.** A unified enforcement point for authentication tokens, rate limiting, schema validation, and payload inspection, increasingly important as containerised and AI-enabled workloads communicate through REST and gRPC APIs. This capability follows the OWASP API Security Top 10 taxonomy [46] and the DevSecOps-for-service-mesh guidance in NIST SP 800-204C [41].

### 4) Layer 4: User and Orchestration-Centric Security

Layer 4 governs identity, access, and policy enforcement across the multi-provider continuum. It implements continuous, context-adaptive authorisation aligned with NIST SP 800-207 [4] and its cloud-native extension SP 800-207A [5]; federated and self-sovereign identity for humans, machines, and autonomous workloads based on the W3C Verifiable Credentials Data Model [47] and, in Europe, the eIDAS 2 framework [48] and its implementing acts for the EU Digital Identity Wallet [49]; and privileged-access controls that reduce reliance on standing credentials. Layer 4 is the layer at which identity-centric and policy-driven controls substitute for assumptions of stable network perimeter control.

**Capability L4.1, Zero-Trust Orchestration.** Policy Enforcement Point and Policy Decision Point architectures that enable continuous authorisation and dynamic access controls

adapted to request context, device posture, workload identity, and operational risk, aligned with NIST SP 800-207 [4] and its cloud-native extension [5].

**Capability L4.2, Federated Identity and Access Management.** Federated and self-sovereign identity mechanisms enabling cryptographically verifiable, decentralised authentication of machines, users, and automated workloads across heterogeneous cloud-edge domains, based on the W3C Verifiable Credentials Data Model [47] and, in Europe, the eIDAS 2 framework [48] and the EU Digital Identity Wallet implementing acts [49].

**Capability L4.3, Privileged Access Management and Just-In-Time Access.** Reduction of over-provisioned standing privileges, a recurring enabler of advanced persistent threat campaigns, by replacing permanent access with time-bounded, contextually justified, and auditable grants, consistent with Zero Trust authorisation principles [4], [5].

### 5) Layer 5: Data-Centric Security

Layer 5 protects data across the continuum lifecycle: at rest, in transit, and in use. It spans confidential computing through Trusted Execution Environments [50]; post-quantum cryptography based on NIST FIPS 203/204/205 [51]–[53]; signalling and data-plane protection; distributed reputation and trust; Quantum Key Distribution for selected high-value links [62], [63]; and Secure Multi-Party Computation for collaborative processing across mutually distrusting parties. Layer 5 is the layer at which GDPR Article 32 [28] and AI Act Article 15 [11] obligations on confidentiality, integrity, and robustness are most directly operationalised.

**Capability L5.1, Hardware-Isolated Confidential Computing.** Protection of data in use through Trusted Execution Environments (TEEs) spanning the cloud-to-IoT data path, including Intel TDX, AMD SEV-SNP, ARM CCA, and the open RISC-V Keystone framework [38], as scoped by the Confidential Computing Consortium [50]. TEEs deliver near-native performance under a hardware-rooted trust assumption, anchored in Capability L1.1 (Hardware Roots of Trust) for remote attestation and providing the execution substrate for privacy-preserving computation (Capability L5.6) when cryptographic-only approaches are infeasible on resource-constrained edge nodes.

**Capability L5.2, Post-Quantum Cryptography.** Migration to quantum-resistant algorithms, namely ML-KEM [51], ML-DSA [52], and SLH-DSA [53], including adaptations for resource-constrained devices exposed to long-term “Harvest Now, Decrypt Later” threats. This capability is complemented by ETSI TS 103 744 hybrid key-exchange profiles [54] and aligned with the NIS CG PQC roadmap [8].

**Capability L5.3, Signalling and Data-Plane Security.** Embedding cryptographic authentication and integrity protection into signalling, control-plane, and data-plane interactions, beyond best-effort packet forwarding, in alignment with 3GPP TS 33.501 [55], O-RAN security specifications [56], [57], and the SNS JU SRIA [58].



**Capability L5.4, Distributed Reputation and Trust Networking.** Distributed reputation and trust mechanisms, potentially supported by Distributed Ledger Technology and smart contracts, enabling decentralised trust evaluation across heterogeneous providers without reliance on a single governance point, and aligned with Simpl federation interfaces [13], [59]–[61].

**Capability L5.5, Quantum Key Distribution for Critical Links.** Information-theoretically secure key exchange over optical fibre for selected point-to-point high-value links, such as government communications and critical infrastructure backbones, complementing algorithmic PQC for link-layer protection and governed by ETSI GS QKD 004 [63] and GS QKD 014 [62]. QKD is treated here as a complement to algorithmic PQC, not a substitute for it.

**Capability L5.6, Privacy-Preserving Computation.** Cryptographic mechanisms enabling computation over confidential data without disclosing inputs to computing parties, comprising (i) Secure Multi-Party Computation (SMPC), allowing mutually distrusting parties to jointly evaluate functions over combined private inputs, with foundations in the threshold-cryptography lineage formalised by NIST [64], and (ii) Homomorphic Encryption (HE), permitting computation directly on ciphertexts under schemes including BFV, BGV, and CKKS, standardised under ISO/IEC 18033-6 [65] and the HomomorphicEncryption.org community specification [66]. The two mechanisms are complementary: SMPC offers cryptographic guarantees at the cost of multi-round communication, while HE provides non-interactive computation on encrypted data at substantial computational overhead ( $10^3$ – $10^6 \times$  slowdown depending on circuit depth and scheme parameters). Together they extend confidential computing (Capability L5.1) beyond single-party workloads to federated analytics across European providers and jurisdictions, satisfying GDPR Articles 25 and 32 on data-protection-by-design and security of processing [28] and AI Act Article 10(5) on appropriate safeguards for processing special-category data in bias monitoring [11].

#### 6) Layer 6: Network Infrastructure and Perimeter-Centric Security

Layer 6 secures the communication fabric of the continuum, including 5G and 6G cores, disaggregated RAN, IT/OT gateways, far-edge compute, and non-terrestrial network segments. It aligns with 3GPP TS 33.501 [55], O-RAN WG11 security specifications [56], Zero Trust for O-RAN [57], the SNS JU SRIA [58], and 3GPP NTN security [67], [68]. It is the layer at which cybersecurity intersects most directly with physical resilience, because failures or compromises in the communication fabric can cascade into operational disruption, safety hazards, or kinetic effects in critical infrastructures [6], [24].

**Capability L6.1, Telco-Cloud and O-RAN Security.** Hardening of distributed 5G and 6G cores and RAN components in alignment with 3GPP TS 33.501 [55], O-RAN WG11 security specifications [56], Zero Trust for O-

RAN [57], and the SNS JU SRIA [58], with emphasis on secure network slice isolation, virtualised RAN protection, and sovereign operational control over critical telecom infrastructure.

**Capability L6.2, Physical Resilience and Safety.** IT/OT gateway security for high-availability edge environments such as smart grids, transport systems, and manufacturing sites, reducing the risk that digital breaches cascade into operational disruption, kinetic damage, or safety hazards [6], [24], [55], [58].

**Capability L6.3, Edge-Native Autonomous Guardians.** Lightweight, localised security control functions, including Security-as-a-Service aggregators, hosted directly at far-edge gateways or within Open-RAN Radio Intelligent Controllers (xApps/rApps) to enable autonomous enforcement at the network periphery [56]–[58].

**Capability L6.4, Non-Terrestrial Network and Satellite Security.** Security requirements for integrating satellite and non-terrestrial network segments into 6G architectures for coverage continuity, backhaul resilience, and emergency operations, aligned with 3GPP NTN specifications [67], [68] and the EU Space Programme.

## B. TRANSVERSAL SECURITY CROSS-LAYERS

The four Transversal Security Cross-Layers define horizontal security capabilities that span the vertical stack and address properties no single layer can provide on its own: lifecycle integrity of software and AI components, operational response capacity, interoperability across providers and jurisdictions, and continuous regulatory assurance.

### 1) Cross-Layer 1: Automated Security Processes and Cyber-Resilient AI Lifecycle

Cross-Layer 1 spans the lifecycle of software and AI component security, from secure development and integration to runtime supply-chain verification and continuous adversarial validation. It treats AI systems both as defensive instruments and as attack targets, the latter framing being required by AI Act Article 9 risk-management obligations for high-risk systems [11]. CL1 aligns with service-mesh DevSecOps patterns [41], the NIST AI Risk Management Framework and its Generative AI Profile [69], [70], the SLSA provenance framework [71], and the adversarial robustness literature [72], with the objective of embedding cyber-resilience into how software and AI components are designed, built, shipped, verified, and tested rather than treating assurance as a post-deployment activity.

**Capability CL1.1, Continuous DevSecOps.** AI-assisted code analysis, vulnerability detection, and remediation integrated into the earliest stages of the software lifecycle, following NIST SP 800-204C [41].

**Capability CL1.2, Model Integrity and DataBOMs.** Protection of AI training and deployment pipelines against data poisoning, unauthorised model manipulation, and insufficient traceability of training data and model artefacts, aligned with the NIST AI Risk Management Framework [69]

and its Generative AI Profile [70], and supporting AI Act Article 9 risk-management requirements [11].

**Capability CL1.3, Dynamic Supply-Chain Verification and Runtime SBOMs.** Dynamic Software Bills of Materials and cryptographic proof-of-execution mechanisms that continuously verify the integrity of containerised microservices and AI components as they migrate across the multi-provider continuum, supported by the SLSA provenance framework [71].

**Capability CL1.4, Automated Adversarial Testing and Continuous Purple Teaming.** Continuous automated penetration testing, coverage-guided and protocol-level fuzzing, and purple-team automation against live or staging environments, providing the adversarial validation counterpart to DevSecOps and grounded in the adversarial robustness literature [72].

## 2) Cross-Layer 2: Dynamic Risk Management and Operational Cyber Resilience

Cross-Layer 2 governs threat detection, analytics, cyber threat intelligence, and operational response across the continuum. It supports SOC federation across Member States [73], integrates AI-native telemetry analytics via the 5G NWDAF architecture [74] and LLM-assisted security operations, and structures the cyber threat intelligence lifecycle using MITRE ATT&CK [76] and STIX/TAXII 2.1 [77], [78] as the structured-CTI foundation, complemented by ENISA threat landscape reporting [75]. Its role is to reduce detection-to-response timelines and improve coordinated operational resilience in environments where human-paced analysis alone is insufficient.

**Capability CL2.1, Automated SOC and Collaborative Threat Intelligence.** Federated SOC and collaborative threat-intelligence mechanisms across Member States, supported by legally protected sharing platforms that reduce institutional trust barriers and enable coordinated situational awareness [73].

**Capability CL2.2, High-Precision Telemetry and Security Signalling.** High-fidelity telemetry and security signalling mechanisms that feed real-time threat-intelligence and detection pipelines while managing bandwidth, latency, and alert-volume constraints, integrating with 5G NWDAF analytics interfaces [74] and threat-landscape baselines [75].

**Capability CL2.3, Internet of Behaviour and Evidence Sharing.** Collection, correlation, and sharing of behavioural evidence across infrastructure components to detect anomalous patterns invisible to purely signature-based defences, structured through MITRE ATT&CK [76] and STIX/TAXII 2.1 [77], [78] and informed by ENISA threat reporting [75].

**Capability CL2.4, AI-Native Analytics Integration (NWDAF and LLMSecOps).** Integration of cybersecurity telemetry with 5G NWDAF analytics [74] and LLM-assisted security operations to support triage, correlation, investigation, and response, framed by the NIST AI RMF and its Generative AI Profile [69], [70].

**Capability CL2.5, Cyber Threat Intelligence Lifecycle Management.** Governance of the CTI lifecycle as a prerequisite for federated SOC operation, using MITRE ATT&CK [76] and STIX/TAXII 2.1 [77], [78], complemented by ENISA threat landscape reporting [75].

**Capability CL2.6, Deception Technology and Active Defence.** Deployment of decoy assets, including honeypots, honeynets, credential canaries, and fake API endpoints, to generate high-fidelity alerts based on adversary interaction and to support earlier detection of lateral movement or reconnaissance [75], [76].

## 3) Cross-Layer 3: Interoperable Security Mechanisms and Open-Source Verifiability

Cross-Layer 3 enables secure interoperation across heterogeneous providers, infrastructures, and jurisdictions. It anchors federation in the Simpl open-source middleware for Common European Data Spaces [59], CAMARA Network APIs for pan-European compute federation [79], semantic interoperability through OASC Minimal Interoperability Mechanisms [60] and the ETSI NGSI-LD API [61], and federated cyber ranges for shared validation infrastructure. Its objective is to convert individually sovereign components into a collectively sovereign European trust fabric while avoiding new forms of vendor lock-in.

**Capability CL3.1, Heterogeneous Interoperability and Simpl Integration.** Alignment with the Simpl open-source smart middleware [59] to support secure federated access, policy portability, and interoperability across Common European Data Spaces, with semantic anchoring in OASC MIMs [60] and the ETSI NGSI-LD API [61].

**Capability CL3.2, Neutral Host and Multi-Tenant Isolation.** Enforcement of Zero Trust edge operations across network partitions, tenants, and shared-infrastructure models, ensuring that infrastructure sharing does not imply implicit trust between operators or workloads [4], [5].

**Capability CL3.3, Pan-European NaaS Federation and Cross-Border Edge Roaming.** Standardised interfaces leveraging CAMARA Network APIs [79] to support a federated Network-as-a-Service model for cross-border edge and connectivity services, in alignment with the Digital Networks Act proposal [80].

**Capability CL3.4, Semantic Security Interoperability.** Common ontologies and data models for security events, asset taxonomies, policy attributes, and risk classifications, operationalised through OASC Minimal Interoperability Mechanisms [60] and the ETSI NGSI-LD API [61].

**Capability CL3.5, Cyber Range and Digital Twin Federation.** Federation of national cyber ranges and digital-twin environments into shared EU-wide validation infrastructure for testing cross-provider security, resilience, and interoperability assumptions, building on EU certification frameworks [13], CONCORDIA outputs [73], and the European industrial cloud-edge roadmap [3], and aligned with the Cyber Solidarity Act.

#### 4) Cross-Layer 4: Compliance-by-Design, Agile Governance and Automated Assurance

Cross-Layer 4 embeds regulatory compliance, governance, and assurance considerations into technical architecture. It operationalises continuous assurance for cloud services (EUCS, when adopted) and edge products (EUCC) [13], [14] through machine-readable control catalogues and assessment results based on NIST OSCAL [81], [82]; cryptographic key management across multi-provider operations; privacy-preserving federated computation satisfying GDPR Article 25 [28]; cyber risk quantification that translates technical posture into financial and operational risk language under NIS2 [9], DORA [12], and NIST CSF 2.0 [34]; and AI trustworthiness and algorithmic accountability for autonomous security decisions under AI Act Article 9 [11], aligned with the NIST AI RMF [69]. Its objective is to shift assurance from point-in-time audit towards continuous, evidence-based, and telemetry-supported verification.

**Capability CL4.1, Post-Quantum Readiness and Zero Trust Architectures.** Continuous Zero Trust verification and post-quantum readiness embedded into cloud-edge infrastructure, aligned with NIST SP 800-207 [4] and SP 800-207A [5] and with FIPS 203/204/205 algorithm selection [51]–[53].

**Capability CL4.2, Continuous Automated Certification.** Lifecycle certification tracking for cloud services (EUCS, when adopted) and edge products (EUCC) [13], [14], operationalised through machine-readable control catalogues, assessment artefacts, and evidence exchange using NIST OSCAL [81], [82].

**Capability CL4.3, Multicloud Key Management as a Service.** Cryptographic key, policy, and lifecycle management across dynamic multi-provider computing-continuum operations, integrating Zero Trust authorisation [4], [5] with PQC algorithm selection [51]–[53].

**Capability CL4.4, Cross-Jurisdictional Federated AI Governance.** Governance and orchestration pattern for AI training and inference over distributed private datasets held across European jurisdictions and providers, combining federated learning with the privacy-preserving computation primitives of Capability L5.6 (SMPC, HE) and the hardware-isolated execution substrate of Capability L5.1 (TEEs) to enforce data minimisation, purpose limitation, and confidentiality at the protocol level rather than through contractual controls alone. This capability addresses the governance, attestation, and audit dimensions—including participant onboarding, model-update provenance, and differential privacy budget accounting—that the underlying primitives do not themselves provide. It supports GDPR Article 25 on data-protection-by-design [28], AI Act Article 10 on data governance for high-risk AI systems [11], and the Data Act's interoperability mandates for cross-border data spaces [27].

**Capability CL4.6, Cyber Risk Quantification and Board-Level Reporting.** Translation of technical security posture into financial, operational, and strategic risk language for decision-makers, aligned with NIS2 governance

obligations [9], DORA [12], and NIST CSF 2.0 governance functions [34].

**Capability CL4.7, AI Trustworthiness and Algorithmic Accountability.** Assurance that AI systems making or supporting autonomous security decisions remain robust, accountable, and auditable, in compliance with AI Act Article 9 [11] and aligned with the NIST AI RMF and Generative AI Profile [69], [70].

## V. STAGE 2 OUTPUT: PRIORITY CLUSTERS AND SEVEN-DIMENSION SYNTHESIS

Stage 2 consolidates the layer and cross-layer capabilities identified in Section IV into three coordinated priority clusters and analyses each through the seven-dimension template defined in Section III. The synthesis here is capability-anchored and time-invariant: it establishes why the priorities cohere, which capabilities they aggregate, which dependencies they share, and which strategic outcomes they target. The temporal projection of each cluster across Phase 1, Phase 2, and Phase 3, including phase-bound activities and Critical Enablers, is presented separately in Section VI, and quantified targets, coverage thresholds, and market-share objectives are concentrated in Section VII, where they can be traced back to specific phase gates and recommendation actions. Where relevant, the analysis below refers to the underlying layer or cross-layer capabilities from which each cluster is derived.

### A. CLUSTER 1: SOVEREIGN TRUSTED INFRASTRUCTURE

Cluster 1 consolidates the capabilities that provide the trusted execution substrate of the continuum: hardware roots of trust and physical-layer assurance (L1.1–L1.2), secure software foundations and update mechanisms (L2.1–L2.2), confidential computing and data-in-use protection (L5.1), post-quantum cryptographic agility and lightweight security for constrained environments (L5.2 and L5.5–L5.6), signalling and data-plane integrity (L5.3), and resilience mechanisms for critical network infrastructure (L6.1–L6.4). Federated policies and autonomous defence gain operational credibility in proportion to the maturity of this execution substrate. The substrate must support hardware integrity through verifiable silicon roots of trust, quantum-resistant confidentiality through crypto-agile schemes aligned with FIPS 203/204/205, and autonomous resilience under network or orchestration-plane compromise. The shared critical dependency is European sovereign fabrication capacity under the EU Chips Act [83], [84].

a: D1 Baseline.

At the hardware and physical layer (L1.1–L1.2), EU-headquartered designers (Infineon, NXP, STMicroelectronics) achieved EUCC “High” for proprietary secure-element families in 2025 [30], but fabrication remains dependent on non-EU foundries and no EU-origin RISC-V processor has received EUCC or Common Criteria certification. At



the software and update layer (L2.1–L2.2), field-deployed edge infrastructure remains dependent on heterogeneous hardening practices and update mechanisms. At the data-centric layer (L5.1–L5.6), edge and IoT environments remain dependent on classical cryptography and non-European TEE families, and extreme-edge devices remain constrained in compute, energy, and memory. The NIS Cooperation Group roadmap [8] provides binding migration milestones for quantum-resistant cryptography. At the network infrastructure layer (L6.1–L6.4), current deployments often assume persistent connectivity and central orchestration.

#### b: D2 High-Impact Activities and Target Outcomes.

Activities aggregate around four target outcomes whose temporal sequencing is the subject of Section VI. The first outcome is European sovereign hardware roots of trust certified to EUCC “High”, advanced through RISC-V secure-element design (L1.1), supported by Chips JU pilot lines, and validated through OpenTitan [37], Keystone [38], and the DARE SGA1 EuroHPC initiative [39]. The second outcome is a quantum-resilient and crypto-agile continuum combining FIPS 203/204/205 [51]–[53] algorithms, ETSI TS 103 744 hybrid key-exchange profiles [54], lightweight cryptography for constrained nodes, confidential computing through hardware-isolated TEEs (L5.1), QKD for selected critical links (L5.5), and privacy-preserving collaborative computation through SMPC and HE (L5.6). The third outcome is resilient field-deployed software and update infrastructure (L2.1–L2.2), grounded in cryptographically verified, failure-resilient update pipelines for edge nodes that are physically remote or otherwise difficult to remediate. The fourth outcome is autonomous Island Mode operation across critical network infrastructure (L6.1–L6.4), aligned with the SNS JU SRIA [58] and validated under 3GPP/ETSI fallback profiles, ensuring that essential services continue to operate during disruption to connectivity or central orchestration. Quantified coverage thresholds, mandatory-deployment perimeters, and procurement-share objectives associated with these outcomes are stated in Section VII.

#### c: D3 Primary Drivers.

The primary threat drivers for this cluster arise from attacks against the trusted execution substrate itself: state-sponsored exploitation of debug interfaces and side-channel leakage in field-deployed nodes (L1.1–L1.2); supply-chain insertion and compromised update paths affecting software and firmware maintenance (L2.1–L2.2); “Harvest Now, Decrypt Later” campaigns targeting long-lived data confidentiality and cryptographic migration gaps (L5.1–L5.6); and cascading failures caused by cyberattacks, extreme weather, or geopolitical disruption to communication and orchestration infrastructure (L6.1–L6.4). The regulatory drivers are correspondingly distributed across the underlying capabilities: CRA Articles 10–11 establish cybersecurity and vulnerability-handling obligations [10]; NIS2 Article 21(2)(e) requires policies and procedures for cryptog-

raphy and encryption [9]; AI Act Articles 9 and 15 create risk-management, accuracy, robustness, and cybersecurity obligations for high-risk AI systems [11]; the NIS CG PQC roadmap defines the migration milestones for quantum-resistant cryptography [8]; DORA Article 11 requires ICT business continuity and disaster recovery capabilities [12]; and the SNS JU SRIA frames resilience as a core requirement for future 6G infrastructures [58].

#### d: D4 Catalysts and Enablers.

The main catalysts for this cluster map directly to the trusted-substrate capabilities. For hardware roots of trust and physical-layer assurance (L1.1–L1.2), the principal enablers are the EU Chips Act pilot lines, including NanoIC, FAMES, APECS, PIXEurope, and WBG [83]; the DARE SGA1 EuroHPC initiative [39]; CEN/CENELEC JTC 13 hardware-security standardisation; and EUCC “High” certification pathways. For secure software foundations and update mechanisms (L2.1–L2.2), relevant enablers include ECCC Strategic Agenda priorities [85], secure update frameworks, and coordinated pilot infrastructures for edge and critical-infrastructure deployments. For post-quantum cryptographic agility, confidential computing, QKD, and privacy-preserving computation (L5.1–L5.6), the primary enablers are NIST FIPS 203/204/205 [51]–[53], ETSI TS 103 744 hybrid key-exchange profiles [54], the NIS CG PQC roadmap, and confidential-computing and data-space pilots. For network resilience and Island Mode capabilities (L6.1–L6.4), SNS JU 6G resilience projects [58], 3GPP/ETSI standardisation, and cross-border 5G/6G testbeds provide the main transition infrastructure.

#### e: D5 Disruptors and Systemic Risks.

The main systemic risks affect different parts of the trusted execution substrate. For hardware roots of trust (L1.1), the principal disruptor is EU Chips Act fabrication-capacity shortfall, with adverse signals reported in the European Court of Auditors review of EU semiconductor strategy [84] reportedly cancelled in 2025, which removed the largest EU-based advanced-node commitment. For physical-layer assurance and constrained edge security (L1.2), energy, memory, and signal-quality constraints may limit deployability of lightweight authentication and fingerprinting techniques, while feature drift and privacy exposure of persistent hardware-level identifiers constrain operational RFF deployment [40]. For secure software foundations and update mechanisms (L2.1–L2.2), the main risks are heterogeneous legacy stacks, incomplete asset inventories, and insufficiently resilient update paths in OT and field-deployed edge environments. For data-centric security (L5.1–L5.6), early arrival of a Cryptographically Relevant Quantum Computer before migration is complete, PQC computational overhead on battery-powered IoT devices, side-channel breakthroughs against current TEE architectures, and incomplete cryptographic inventories of legacy OT/IoT systems could delay transition. For network resilience (L6.1–L6.4), dependencies



on central orchestration, telecom supply-chain concentration, and insufficiently validated fallback modes could weaken Island Mode readiness.

#### f: D6 Failsafe Mechanisms.

Failsafe mechanisms should preserve security and continuity when parts of the trusted substrate are delayed, compromised, or only partially available. For hardware and physical-layer capabilities (L1.1–L1.2), these include redundant trust anchors, layered lightweight cryptography, PHY signatures as last-resort authentication under energy starvation, and tamper-resistant local decision logic. For secure software foundations and update mechanisms (L2.1–L2.2), the main failsafes are staged rollout, rollback, secure recovery partitions, and autonomous node quarantine via eBPF network policy when attestation or update verification fails. For data-centric security (L5.1–L5.6), the relevant failsafes are TEE abstraction layers to avoid vendor lock-in during the dependency period, hybrid cryptographic schemes combining classical and PQC mechanisms for backward compatibility and harvest resistance, crypto-agility frameworks enabling rapid algorithm replacement, and diversification across PQC algorithm families, including lattice-, code-, and hash-based approaches, to avoid single-point cryptanalytic failure. For network resilience and Island Mode (L6.1–L6.4), failsafes include segmented autonomy zones, redundant local control paths, fallback routing, and validated autonomous operation for critical services when connectivity or orchestration planes are degraded.

#### g: D7 Transversal Meta-Layer.

**Sovereignty:** Across L1, L2, L5, and L6, the cluster reduces dependence on proprietary hardware roots of trust, non-European TEE stacks, classical cryptographic assumptions, and externally controlled network resilience mechanisms. Hardware sovereignty is primarily addressed through EU-designed secure elements and EUCC “High” assurance (L1.1), while cryptographic sovereignty is addressed through PQC migration, crypto-agility, and selected critical-link QKD (L5.2 and L5.5). **Trust and Ethics:** EUCC “High” aligns hardware assurance with Common Criteria EAL4+ and AVA\_VAN.5 expectations; PQC and confidential-computing migration support GDPR Article 32 confidentiality and integrity obligations [28]; and the 18-Member-State joint statement “Securing Tomorrow, Today” (November 2024) formalises PQC as a collective sovereignty concern. **Sustainable Security:** Energy-efficient secure silicon and lightweight cryptography support far-edge deployment without premature device replacement, while resilient update mechanisms, crypto-agility, and validated Island Mode extend the operational lifetime of critical infrastructure assets.

## B. CLUSTER 2: COGNITIVE SECURITY AND AUTONOMOUS CYBER DEFENCE

Cluster 2 consolidates the capabilities that enable cognitive security operations and autonomous cyber defence across

the continuum: software and AI lifecycle assurance (CL1.2–CL1.4), federated SOC and cyber threat intelligence functions (CL2.1 and CL2.5), high-precision telemetry and AI-native analytics (CL2.2–CL2.4), deception and active defence (CL2.6), Zero Trust reconfiguration and containment (L4.1), telecom telemetry integration (L6.1), and trustworthy algorithmic accountability for autonomous security decisions (CL4.7). The cluster is structured by three inseparable sub-dimensions: AI as a security instrument, including sovereign models, edge-native agents, autonomous SOC, and Moving Target Defence; AI as an adversary amplifier, including polymorphic malware, automated exploit discovery, and scaled social engineering; and AI as a target, where training pipelines and inference endpoints become attack surfaces under AI Act Article 9.

#### a: D1 Baseline.

At the operational cyber-resilience layer (CL2.1–CL2.6), European SOC and CTI capabilities remain fragmented across organisations, Member States, and commercial platforms, with US-platform SIEMs dominating the EU enterprise market and no EU-headquartered vendor in the Gartner Leaders quadrant [32]. At the AI lifecycle layer (CL1.2 and CL1.4), sovereign security-specific foundation models and standardised adversarial robustness validation for security AI remain underdeveloped; OpenEuroLLM [86] does not yet include a dedicated security stream. At the analytics and telemetry layer (CL2.2–CL2.4 and L6.1), NWDAF-enabled telecom telemetry and LLM-assisted SecOps remain at pilot or early integration stage rather than being operationally federated across the continuum. At the governance and assurance layer (CL4.7), trustworthy, auditable, and AI Act-compliant autonomous security decision-making remains immature, particularly for systems that support containment, reconfiguration, or prioritisation of critical incidents.

#### b: D2 High-Impact Activities and Target Outcomes.

Activities aggregate around four target outcomes whose temporal sequencing is the subject of Section VI. The first outcome is sovereign cognitive security analytics, advanced through EU-trained security-focused small language models on EuroHPC AI Factory compute [87] and edge-deployable inference capable of operating without external cloud dependency (CL2.4). The second outcome is federated SOC and structured CTI operation, supported by MITRE ATT&CK [76], STIX/TAXII 2.1 [77], [78], and ENISA threat-landscape baselines [75], with NWDAF telemetry pipelines [74] integrated into AI-assisted triage and investigation (CL2.1, CL2.2–CL2.5, and L6.1). The third outcome is bounded autonomous response, comprising Zero-Trust reconfiguration aligned with NIST SP 800-207 [4] (L4.1), policy-constrained autonomous triage under ETSI GS ZSM 009 [88] (CL2.4), Moving Target Defence and deception infrastructure (CL2.6 and L4.1), and adversarial-robustness validation for security AI (CL1.4) operating under AI Act Article 9 conformity assessment [11]. The fourth

outcome is trustworthy and accountable autonomous security decision-making aligned with the NIST AI RMF and Generative AI Profile [69], [70] (CL4.7). Quantified dwell-time targets, sovereign-tool coverage objectives, and the perimeter of mandatory autonomous-agent certification are stated in Section VII.

#### c: D3 Primary Drivers.

The primary threat drivers for this cluster arise from the accelerating use of AI by both defenders and adversaries. At the AI lifecycle and model-assurance layer (CL1.2–CL1.4), generative AI can be weaponised for polymorphic malware, automated exploit discovery, adversarial prompt manipulation, and scaled social engineering, while training pipelines and inference endpoints become attack surfaces in their own right. At the operational cyber-resilience layer (CL2.1–CL2.6), nation-state actors increasingly blend espionage with organised crime, use memory-only execution, abuse legitimate administrative tools, and exploit fragmented SOC and CTI workflows to extend dwell time. At the analytics and telecom-telemetry layer (CL2.2–CL2.4 and L6.1), high-volume telemetry, NWDAF integration, and LLM-assisted triage create both detection opportunities and new risks of automation bias, data leakage, and adversarial manipulation. At the governance and assurance layer (CL4.7), autonomous or semi-autonomous security decisions raise accountability, auditability, and conformity-assessment challenges. The regulatory drivers are correspondingly concentrated around AI and operational resilience: AI Act Articles 9, 13, and 15 create mandatory requirements for high-risk autonomous security AI [11]; NIS2 Article 21(2)(b) establishes incident-handling obligations [9]; and conformity-assessment capacity is expected to lag obligations significantly [17].

#### d: D4 Catalysts and Enablers.

For AI lifecycle assurance and security-specific models (CL1.2–CL1.4), EuroHPC AI Factories provide the principal compute and training infrastructure, with cybersecurity emerging as a vertical in several sites [87]. For federated SOC, CTI, and operational response (CL2.1 and CL2.5), MITRE ATT&CK [76], STIX/TAXII 2.1 [77], [78], ENISA threat-landscape reporting [75], and the ECCC Strategic Agenda [85] provide the main knowledge, coordination, and governance enablers. For AI-native analytics, NWDAF integration, and autonomous triage (CL2.2–CL2.4 and L6.1), ETSI GS ZSM 009 [88], telecom analytics interfaces, and 5G/6G testbeds provide the operational automation base. For trustworthy autonomous security decisions (CL4.7), the NIST AI RMF and Generative AI Profile [69], [70], together with AI Act conformity-assessment infrastructure, provide the principal assurance framework.

#### e: D5 Disruptors and Systemic Risks.

The main systemic risks affect both the AI components and the cyber-defence workflows that depend on them. For AI lifecycle assurance (CL1.2–CL1.4), hallucinations, adversar-

ial prompt manipulation, data leakage into models, poisoned training data, and the absence of a standardised adversarial robustness test suite for security AI could undermine trust in autonomous or semi-autonomous tools. For operational cyber resilience (CL2.1–CL2.6), intelligence fragmentation, weak information-sharing incentives, and adversarial adaptation outpacing defensive innovation could prevent federated SOCs from achieving the intended dwell-time reduction. For analytics and telecom telemetry (CL2.2–CL2.4 and L6.1), high-volume data flows may increase false positives, automation bias, and dependence on opaque model outputs unless explainability and auditability are built into triage and response workflows. For trustworthy AI governance (CL4.7), conformity-assessment body capacity shortfalls [17] could delay deployment of compliant autonomous security agents even where the technical capabilities are mature.

#### f: D6 Failsafe Mechanisms.

Failsafe mechanisms should bound autonomy and preserve operational control when AI-assisted security systems behave unpredictably or lack sufficient assurance. For AI lifecycle capabilities (CL1.2–CL1.4), failsafes include adversarial robustness testing, red-teaming, model rollback, dataset provenance checks, and shadow-mode operation before production deployment. For operational cyber-resilience capabilities (CL2.1–CL2.6), layered detection, deception, rapid recovery playbooks, and human-in-the-loop validation for Tier-1 critical actions should prevent over-reliance on automated triage. For autonomous analytics and orchestration (CL2.4, L4.1, and L6.1), bounded autonomy governed by ETSI GS ZSM 009 [88], policy-constrained orchestration, audit logging, and shadow-mode rollback should limit unstable control loops. For Zero Trust containment and accountable decision-making (L4.1 and CL4.7), autonomous real-time micro-segmentation should quarantine compromised components only within predefined policy boundaries, with traceable justification and escalation paths.

#### g: D7 Transversal Meta-Layer.

**Sovereignty:** Across CL1, CL2, CL4.7, L4.1, and L6.1, the cluster reduces dependence on non-European SIEM platforms, external foundation-model providers, and opaque cloud-based inference for security operations by promoting EU-sovereign security models, edge-native analytics, federated SOC coordination, and auditable autonomous response. **Trust and Ethics:** AI Act Article 9 compliance-by-design, explainability, auditability, and accountability are central for autonomous or semi-autonomous security decisions, while NIS2 sector-specific implementing acts should reference structured CTI frameworks such as MITRE ATT&CK [76] to improve operational comparability across sectors. **Sustainable Security:** Edge-native SLMs and localised analytics can reduce inference-time energy costs and latency relative to repeated cloud round-trips for high-volume telemetry, while automation can reduce analyst overload if bounded by human oversight, robust testing, and transparent governance.

### C. CLUSTER 3: FEDERATED OPERATIONAL SOVEREIGNTY

Cluster 3 consolidates the capabilities that transform sovereign components into a collectively sovereign continuum: heterogeneous interoperability and Simpl integration (CL3.1), neutral-host and multi-tenant isolation (CL3.2), pan-European NaaS federation and cross-border edge roaming (CL3.3), semantic security interoperability (CL3.4), cyber range and digital twin federation (CL3.5), dynamic supply-chain verification and Runtime SBOMs (CL1.3), continuous automated certification and assurance (CL4.2), privacy-preserving federated computation (CL4.4), and Zero Trust policy enforcement across shared infrastructure (L4.1). Supply-chain verification bridges Clusters 1 and 3 because software, runtime, and federation assurance increasingly depend on the hardware roots of trust and attestation capabilities matured in Cluster 1.

#### a: D1 Baseline.

At the interoperability layer (CL3.1–CL3.4), European cloud-edge ecosystems remain fragmented across proprietary vendor silos, incompatible security policies, and uneven semantic models. The Simpl-Open iterative MVP has been available since January 2025 [59], but OASC MIM6 (Security management) remains a “Work item” without a formally published specification. At the supply-chain and runtime-verification layer (CL1.3 and CL4.2), current supply chains remain opaque and globally distributed: CRA Annex I Part II mandates static SBOM disclosure [10], NIS2 Article 21(2)(h) mandates supply-chain risk assessment [9], and DORA Article 28 requires continuous ICT third-party monitoring [12], operationalised through the first Critical ICT Third-Party Provider designations in November 2025 [89]. At the federation and connectivity layer (CL3.3, L6.1, and L4.1), the CAMARA Network APIs programme [79] and GSMA Open Gateway [90] provide growing API surface, but cross-border edge compute roaming does not yet exist at operational scale. At the governance and assurance layer (CL4.2–CL4.6), continuous assurance, cyber risk quantification, and policy portability remain insufficiently integrated across providers and jurisdictions, while hyperscaler concentration in the EU cloud market remains structurally significant [7].

#### b: D2 High-Impact Activities and Target Outcomes.

Activities aggregate around four target outcomes whose temporal sequencing is the subject of Section VI. The first outcome is procurement-enforceable security interoperability through Simpl, in which OASC MIM6 progresses from work item to formally published specification, then to ratification as a CEN/CENELEC standard referenced in the EU Cloud Rulebook (CL3.1 and CL3.4). The second outcome is runtime supply-chain attestation, advanced from CRA-mandated static SBOM disclosure (CL1.3 and CL4.2) towards Runtime SBOMs supported by cryptographic proof-of-execution and zero-knowledge-proof batch verification at transaction scale. The third outcome is operational pan-

European NaaS federation, deployed through harmonised CAMARA APIs across bilateral 5G corridor agreements and scaling to a federated Network-as-a-Service model for cross-border edge and connectivity services (CL3.3 and L6.1), with Zero Trust isolation enforced across shared and multi-tenant infrastructure (CL3.2 and L4.1). The fourth outcome is cross-jurisdictional federated AI governance through “Data Visa” mechanisms enabling dynamic AI model migration across European jurisdictions under privacy-preserving computation primitives (CL3.1, CL4.4, and CL4.7). Quantified procurement-share thresholds, mandatory-deployment perimeters, and cross-border roaming scale objectives are stated in Section VII.

#### c: D3 Primary Drivers.

The primary threat drivers for this cluster arise from the difficulty of federating security across heterogeneous providers, jurisdictions, and supply chains. At the interoperability and federation layer (CL3.1–CL3.4), proprietary security policies, incompatible APIs, and uneven semantic models create fragmentation that prevents European operators from scaling against US and Chinese hyperscalers. At the supply-chain and runtime-verification layer (CL1.3 and CL4.2), SolarWinds-class software supply-chain attacks demonstrate the risk of trusting static artefacts, opaque build pipelines, and unverifiable runtime components. At the shared-infrastructure and network-federation layer (CL3.2, CL3.3, L4.1, and L6.1), disaggregated telco clouds and cross-border edge deployments introduce multi-vendor complexity, tenant-isolation risks, and policy-enforcement gaps. The regulatory drivers are correspondingly distributed across the underlying capabilities: the Data Act strengthens portability and switching obligations [27]; NIS2 requires coordinated incident response and supply-chain risk management [9]; the CRA introduces SBOM obligations [10]; DORA requires continuous ICT third-party monitoring [12]; and the Digital Networks Act proposal [80] signals EC intent for network federation, although it does not mandate specific API standards.

#### d: D4 Catalysts and Enablers.

For heterogeneous interoperability and Simpl integration (CL3.1 and CL3.4), Digital Europe Programme funding for Simpl [59], CEN-CLC/JTC 25 data-space standardisation, the Eclipse Dataspace Protocol, and OASC/NGSI-LD semantic interoperability provide the main standardisation and implementation base. For pan-European NaaS federation and cross-border edge roaming (CL3.3 and L6.1), SNS JU activities [58], CAMARA APIs, the Digital Networks Act proposal [80], and emerging aggregator models provide the principal technical and ecosystem enablers. For supply-chain verification and continuous assurance (CL1.3 and CL4.2), the IPCEI-CIS ICRA Reference Architecture [91], the EU Ecodesign Regulation Digital Product Passport, and DORA Delegated Regulation (EU) 2024/1502 for CTPP assessment [12] provide the principal policy and assurance in-

struments. For Zero Trust isolation and shared-infrastructure governance (CL3.2 and L4.1), reusable policy profiles, certification evidence exchange, and multi-tenant isolation pilots are the key implementation enablers.

e: D5 Disruptors and Systemic Risks.

For interoperability and semantic security (CL3.1 and CL3.4), vendor resistance to open standards, proprietary API ecosystems, and uneven Member State compliance maturity could prevent convergence on common security mechanisms. For neutral-host and multi-tenant isolation (CL3.2 and L4.1), weakest-link failures in federated systems could allow one poorly governed participant to undermine the trust assumptions of the wider federation. For supply-chain verification and continuous assurance (CL1.3 and CL4.2), limited multi-tier auditability, concentration of manufacturing and software dependencies, and SME exclusion from compliance pathways could weaken adoption. For NaaS federation and cross-border edge roaming (CL3.3 and L6.1), hyperscaler platform dominance through proprietary API ecosystems [7] could limit European scale-up. Across all federation capabilities, unresolved US CLOUD Act jurisdictional conflict [29] remains a structural sovereignty risk.

f: D6 Failsafe Mechanisms.

Failsafe mechanisms should preserve trust when federation assumptions fail or when some providers cannot meet the shared baseline. For heterogeneous interoperability and Simpl-based federation (CL3.1 and CL3.4), failsafes include autonomous unfederation, policy rollback, and fallback to minimum common security profiles. For neutral-host and multi-tenant isolation (CL3.2 and L4.1), Zero Trust architectures should ensure that federation does not imply implicit trust [4], with cryptographic attestation of policy enforcement and graceful isolation of unverified components into quarantined network slices. For supply-chain verification and continuous assurance (CL1.3 and CL4.2), runtime attestation, trusted fallback components, and multi-vendor diversification reduce dependency on any single provider or assurance path. For NaaS federation and network resilience (CL3.3 and L6.1), graceful degradation should redirect workloads to national infrastructure, regional autonomy zones, or multi-operator fallback routing when cross-border federation is unavailable or legally constrained.

g: D7 Transversal Meta-Layer.

**Sovereignty:** Across CL3, CL1.3, CL4.2, L4.1, and L6.1, the cluster reduces hyperscaler lock-in by making interoperability, runtime assurance, policy portability, and cross-border edge federation enforceable rather than aspirational, while also addressing structural CLOUD Act conflict. **Trust and Ethics:** Transparent and verifiable federation mechanisms support cross-border trust by making policy enforcement, runtime integrity, and assurance evidence auditable across providers and jurisdictions, with national procurement signalling such as the 2025 DINUM Bilan [92] demonstrating

that sovereign-leaning procurement is operationally feasible. **Sustainable Security:** Shared federated infrastructure, reusable certification evidence, and common security profiles reduce duplication of compute, audit, and certification overhead across Member States while supporting SME access to compliant cloud-edge services.

## VI. STAGE 3 OUTPUT: PHASE LOGIC AND CLUSTER TRANSITION MATRICES

Stage 3 projects the three priority clusters of Section V onto the 2026–2040 horizon. The capability-anchored synthesis of Stage 2 is converted into a temporal trajectory: where each cluster sits in each phase, which observable conditions characterise that phase, and which single gate condition, the Critical Enabler, must be delivered for the cluster to progress to the next phase. Tables 2–4 present the per-cluster transition matrices. Each matrix arranges the four assessment perspectives of Section III-C (R&I maturity; policy, regulation, and standards; operational deployment and adoption; ecosystem uptake and strategic impact) as rows, with the three phases as columns, and uses two scalar bands (Operational Deployment Phase and Strategic Sovereignty Level) to position the cluster within each phase column. Each phase column terminates in a Critical Enabler row. The narrative subsections that follow walk through what changes between phases and which gate condition controls the change, rather than restating the static cluster analysis. Cross-cutting dependencies shared by two or more clusters are addressed in Section VII-B, and quantified coverage thresholds, market-share targets, and dwell-time objectives are concentrated in Section VII, where each numeric target is traced to the phase gate it depends on.

### A. CLUSTER 1 TRAJECTORY: SOVEREIGN TRUSTED INFRASTRUCTURE

Phase 1 establishes the technical and certification preconditions for a sovereign trusted execution substrate. Hardware roots of trust (L1.1) and resilient-by-design infrastructure (L6.1–L6.4) advance through Horizon Europe Cluster 3 consortia targeting EU-origin RISC-V secure elements for EUCC “High” pre-evaluation, with OpenTitan [37], Keystone [38], and the DARE SGA1 EuroHPC initiative [39] as principal vehicles, and with Island Mode soft-state protocols validated in 5G testbeds aligned with the SNS JU SRIA [58]. Post-quantum cryptographic agility, lightweight cryptography, and confidential computing (L5.1–L5.6) remain in applied research, with FIPS 203/204/205 [51]–[53] implementations optimised for constrained microcontrollers and hybrid ML-KEM plus ECDH-P384 under ETSI TS 103 744 [54] deployed in cloud-side software, while far-edge IoT continues to operate predominantly under classical cryptography. The regulatory perimeter is binding throughout this phase, with NIS2 Article 21(2)(e) cryptographic-policy obligations [9], CRA Articles 10–11 hardware-security requirements [10], AI Act Articles 9 and 15 hardware-integrity obligations [11], and the NIS Cooperation Group migration milestones [8]



**Table 2.** Cluster 1 transition matrix: Sovereign Trusted Infrastructure.

|                                              | Phase 1 (2026–2030)                                                                                                                                                                                                                                                                                                                                                           | Phase 2 (2030–2035)                                                                                                                                                                                                                                                                                                | Phase 3 (2035–2040)                                                                                                                                                                                                                                                                                      |
|----------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>ODP</b>                                   | Applied R&D → Fragmented Piloting                                                                                                                                                                                                                                                                                                                                             | Standardised Integration                                                                                                                                                                                                                                                                                           | Ubiquitous Operation                                                                                                                                                                                                                                                                                     |
| <b>SSL</b>                                   | High Foreign Dependency                                                                                                                                                                                                                                                                                                                                                       | Emerging EU Ecosystem                                                                                                                                                                                                                                                                                              | Full Sovereign Autonomy                                                                                                                                                                                                                                                                                  |
| <b>R&amp;I Maturity</b>                      | EU-origin RISC-V secure elements (OpenTitan, Keystone, DARE SGA1) target EUCC “High” pre-evaluation; FIPS 203/204/205 implementations optimised for constrained MCUs; hybrid ML-KEM + ECDH-P384 under ETSI TS 103 744 deployed cloud-side, with far-edge IoT predominantly classical; Island Mode soft-state protocols validated in 5G testbeds aligned with the SNS JU SRIA. | RISC-V Keystone TEEs deployed in multi-provider Data Spaces alongside Intel TDX and AMD SEV-SNP, with cross-vendor attestation interoperability via TEE abstraction layers; lightweight PQC demonstrated on zero-energy nodes in 6G ISAC testbeds.                                                                 | Privacy-preserving computation operates end-to-end from IoT sensor to HPC compute node across federated Data Spaces; zero-energy physical-layer authentication via RFF supports battery-less mMTC identity; swarm-based agentic AI enable autonomous cyber-physical recovery without cloud intervention. |
| <b>Policy, Regulation, and Standards</b>     | NIS2 Art 21(2)(e), CRA Arts 10–11, AI Act Arts 9 and 15, and the Digital Networks Act proposal [80] signals Commission intent without mandating API standards, and NIS CG PQC migration milestones apply concurrently; EUCC “High” framework operational.                                                                                                                     | Hybrid PQC under ETSI TS 103 744 becomes procurement default for new Industrial IoT and DORA-scoped financial-sector deployments, consistent with the NIS CG PQC milestone for high-risk critical infrastructure; autonomous Island Mode certified under 3GPP/ETSI fallback profiles in cross-border 5G corridors. | EUCC “High” mandatory for new EU ICT products without waiver; PQC-native silicon becomes EU procurement default; continuous automated EUCC lifecycle maintenance replaces point-in-time audit.                                                                                                           |
| <b>Operational Deployment and Adoption</b>   | Operational TEE deployments dominated by Intel TDX, AMD SEV-SNP, and Arm TrustZone/CCA; EUCC “High” achieved for proprietary EU secure-element families (Infineon, NXP, ST) in 2025; no EU-origin RISC-V processor EUCC- or CC-certified; Chips JU pilot lines reach pre-production rather than commercial-source milestones.                                                 | First EUCC “High” certified EU-manufactured RISC-V secure elements enter production from Chips JU pilot lines and are integrated into critical-infrastructure procurement (energy substations, telecommunications) across multiple Member States.                                                                  | EU-designed RISC-V secure elements become the hardware default for new EU ICT products; zero-energy PHY authentication operates as a standard identity mechanism in battery-less mMTC deployments.                                                                                                       |
| <b>Ecosystem Uptake and Strategic Impact</b> | Fabrication remains dependent on non-EU foundries; sovereignty posture is defensive (defined by reducing dependency)                                                                                                                                                                                                                                                          | EU-designed secure elements transition from absent to credible alternative in regulated-sector procurement; foreign TPMs cease to be the sole option; second-source agreements between EU foundries provide production resilience.                                                                                 | Strategic posture transitions from defensive sovereignty to offensive leadership, with adoption of at least one EU hardware-security standard within ISO/IEC or ITU-T as the evidence test for the leadership transition.                                                                                |
| <b>Critical Enabler</b>                      | EUCC “High” pre-evaluation path for an EU-origin, auditable RISC-V secure element, coupled with Chips JU pilot-line access and TEE abstraction layers to mitigate continued lock-in.                                                                                                                                                                                          | Production availability of EU-manufactured, EUCC “High” RISC-V secure elements with second-source fabrication resilience and cross-vendor attestation interoperability.                                                                                                                                            | Mandatory lifecycle assurance for EUCC “High” hardware, PQC-native procurement, and international adoption of at least one EU-origin hardware-security standard.                                                                                                                                         |

applying concurrently. Operational TEE deployments remain dominated by Intel TDX, AMD SEV-SNP, and Arm TrustZone/CCA, and no EU-origin RISC-V processor is EUCC- or Common Criteria-certified at the start of the phase [31]. EU-headquartered designers achieved EUCC “High” for proprietary secure-element families in 2025 [30], but fabrication remains dependent on non-EU foundries and the Chips JU pilot lines [83] reach pre-production rather than commercial-source milestones.

Phase 2 industrialises the Phase 1 pipeline. The first EUCC “High” certified EU-manufactured RISC-V secure elements enter production from Chips JU pilot lines and are integrated into critical-infrastructure procurement in energy-substation and telecommunications applications across multiple Member States. RISC-V Keystone TEEs are deployed in multi-provider Data Spaces alongside Intel TDX and

AMD SEV-SNP, with cross-vendor attestation interoperability ensured through the TEE abstraction layers established in Phase 1. Hybrid PQC under ETSI TS 103 744 [54] becomes the procurement default for new Industrial IoT and DORA-scoped financial-sector deployments [12], consistent with the NIS CG migration milestone for high-risk critical infrastructure [8]; lightweight PQC is demonstrated on zero-energy nodes in 6G Integrated Sensing and Communication testbeds; and autonomous Island Mode is certified under 3GPP/ETSI fallback profiles in cross-border 5G corridors. The strategic shift in this phase is procurement-side: EU-designed secure elements transition from absent in regulated-sector procurement to a credible alternative, while foreign TPMs cease to be the sole option for new regulated-sector deployments and second-source agreements between EU foundries provide production resilience.

Phase 3 consolidates the trusted execution substrate as the default operating model. EU-designed RISC-V secure elements become the hardware default for new EU ICT products; privacy-preserving computation operates end-to-end from IoT sensor to HPC compute node across federated Data Spaces; zero-energy physical-layer authentication via RF fingerprinting (L1.2) becomes a standard identity mechanism for battery-less mMTC devices, deployed under privacy-preserving pipelines that address the feature-drift and persistent-identifier exposure constraints documented in the operational RFF literature [40]; and swarm-based agentic AI enable autonomous cyber-physical recovery without cloud intervention. EUCC “High” becomes mandatory for new EU ICT products without waiver, PQC-native silicon becomes the EU procurement default, and continuous automated EUCC lifecycle maintenance replaces point-in-time audit. The strategic posture transitions from defensive sovereignty (defined by reducing dependency) to offensive leadership (defined by setting the terms of global hardware-security competition), with adoption of at least one EU hardware-security standard within ISO/IEC or ITU-T treated as the evidence test for the leadership transition.

### **B. CLUSTER 2 TRAJECTORY: COGNITIVE SECURITY AND AUTONOMOUS CYBER DEFENCE**

Phase 1 establishes the technical and assurance preconditions for cognitive cyber defence under conditions of structural dependency. Cognitive security operations advance through commercial LLMs integrated into SOC analyst workflows for alert triage and MITRE ATT&CK technique mapping [76], NWDAF telemetry pipelines [74] routed into centralised AI analytics in 5G SA pilot deployments, and ML-based User and Entity Behaviour Analytics applied to APT IoC detection. The phase target is a credible demonstration of dwell-time reduction in at least one certified pilot in a NIS2-essential entity, against the European baseline. AI-as-target protections lag operationally: adversarial-robustness testing methodologies advance from academic research towards pilot readiness, but no operational certification framework for security-specific AI agents exists at the start of the phase, opening a regulatory gap relative to AI Act Articles 9, 13, and 15 [11], [17]. Operational deployment continues to depend on US-platform LLMs as analyst copilots and on rule-based SIEMs as the primary detection infrastructure across regulated sectors, with no certified autonomous containment agent reaching deployment. The market position is one of structural dependency [32], and a first EU-sovereign security-focused small language model only begins training on EuroHPC AI Factory compute [87] during this phase.

Phase 2 operationalises sovereign cognitive defence. EU-sovereign small language models trained on MITRE ATT&CK-aligned corpora are deployed at the edge for real-time SOC analytics without cloud-inference dependency; Guardian nodes execute autonomous Zero-Trust reconfigurations consistent with NIST SP 800-207 [4] under human-in-the-loop escalation for Tier-1 critical

actions; and autonomous incident triage operates under ETSI GS ZSM 009 governance [88], with measurable dwell-time reductions in certified DORA-compliant, energy, and telecommunications deployments across multiple Member States. Adversarial-robustness validation pipelines become mandatory under AI Act Article 9 conformity assessment, supported by a CEN/CENELEC- or ETSI TC CYBER-published test suite for security AI; deception infrastructure (L6 honeynets, credential canaries, fake API endpoints) is integrated into the autonomous detection loop to provide continuous model calibration; NWDAF-integrated analytics become standard in new 5G SA deployments; and EuroHPC AI Factory compute contracts dedicated to EU-sovereign security-model training are activated. The strategic shift in this phase is the emergence of certified EU-sovereign alternatives in regulated-sector SOC procurement, and the first export of EU AI security tools to partner jurisdictions.

Phase 3 establishes autonomous cyber defence as the operational baseline. Swarm-based agentic AI defends the continuum through self-evolving decentralised threat intelligence without central-coordination dependency; Moving Target Defence continuously shuffles IP addressing, network topology, and software diversity, suppressing APT reconnaissance success in certified environments; adversarial-robustness validation is fully automated, with no security AI agent operating in EU critical infrastructure without an active conformity certification; and human intervention is reserved for exceptional legal escalation. Dwell-time targets converge on the threshold defined in Section VII. Dwell-time targets converge on the threshold defined in Section VII. As in Cluster 1, the Phase 3 transition is evidence-tested through external adoption: AI Act-derived frameworks for trustworthy autonomous security agents are referenced in bilateral digital-partnership agreements, the EU AI Office and ENISA jointly contribute to the international standard for security-AI accountability, and the formal adoption of EU security-AI governance by at least one non-EU jurisdiction serves as the evidence test for the leadership transition.

### **C. CLUSTER 3 TRAJECTORY: FEDERATED OPERATIONAL SOVEREIGNTY**

Phase 1 establishes the technical and standards preconditions for federation under conditions of regulatory fragmentation. Security interoperability advances through Simpl security MIM proof-of-concepts in Common European Data Spaces, including Manufacturing, Health [93], and Energy [59]–[61], with manual policy coordination and OASC MIM6 (Security management) progressing from work-item status towards formally published specification. Pan-European NaaS federation reaches bilateral interconnection: harmonised CAMARA Network APIs [79] are deployed in bilateral 5G corridor agreements, but cross-border edge-compute roaming does not yet exist at operational scale. Supply-chain verification remains at static disclosure under CRA Annex I Part II [10], complemented by NIS2 Article 21(2)(h) supply-chain risk assessment [9] and DORA Ar-

**Table 3.** Cluster 2 transition matrix: Cognitive Security and Autonomous Cyber Defence.

|                                              | Phase 1 (2026–2030)                                                                                                                                                                                                                                                                                                                                                                                                             | Phase 2 (2030–2035)                                                                                                                                                                                                                                                                                                                                                                   | Phase 3 (2035–2040)                                                                                                                                                                                                                                |
|----------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>ODP</b>                                   | Applied R&D → Fragmented Piloting                                                                                                                                                                                                                                                                                                                                                                                               | Standardised Integration                                                                                                                                                                                                                                                                                                                                                              | Ubiquitous Operation                                                                                                                                                                                                                               |
| <b>SSL</b>                                   | High Foreign Dependency                                                                                                                                                                                                                                                                                                                                                                                                         | Emerging EU Ecosystem                                                                                                                                                                                                                                                                                                                                                                 | Full Sovereign Autonomy                                                                                                                                                                                                                            |
| <b>R&amp;I Maturity</b>                      | Commercial LLMs integrated into SOC analyst workflows for alert triage and MITRE ATT&CK technique mapping; NWDAF telemetry pipelines routed into centralised AI analytics in 5G SA pilots; ML-based UEBA applied to APT IoC detection; adversarial-robustness testing methodologies advance from academic research to pilot readiness; a first EU-sovereign security-focused SLM begins training on EuroHPC AI Factory compute. | EU-sovereign SLMs trained on MITRE ATT&CK-aligned corpora deployed at the edge for real-time SOC analytics without cloud-inference dependency; Guardian nodes execute autonomous Zero-Trust reconfigurations under human-in-the-loop escalation for Tier-1 critical actions; deception infrastructure integrated into the autonomous detection loop for continuous model calibration. | Swarm-based agentic AI defends the continuum through self-evolving decentralised threat intelligence without central-coordination dependency; Moving Target Defence continuously shuffles IP addressing, network topology, and software diversity. |
| <b>Policy, Regulation, and Standards</b>     | AI Act Arts 9, 13, and 15 obligations active, but no operational certification framework for security-specific AI agents at the start of the phase, opening a regulatory gap; conformity-assessment capacity expected to lag obligations.                                                                                                                                                                                       | Adversarial-robustness validation pipelines mandatory under AI Act Art 9 conformity assessment, supported by a CEN/CENELEC- or ETSI TC CYBER-published test suite for security AI; autonomous incident triage operates under ETSI GS ZSM 009 governance.                                                                                                                              | Adversarial-robustness validation fully automated; no security AI agent operates in EU critical infrastructure without active conformity certification; human intervention reserved for exceptional legal escalation.                              |
| <b>Operational Deployment and Adoption</b>   | US-platform LLMs serve as analyst copilots and rule-based SIEMs as primary detection infrastructure across regulated sectors; no certified autonomous containment agent in deployment; phase target is credible dwell-time reduction in at least one certified pilot in a NIS2-essential entity, against the European baseline.                                                                                                 | Measurable dwell-time reductions in certified DORA-compliant, energy, and telecommunications deployments across multiple Member States; NWDAF-integrated analytics become standard in new 5G SA deployments; EuroHPC AI Factory compute contracts dedicated to EU-sovereign security-model training are activated.                                                                    | Dwell-time targets converge on the threshold defined in Section VII; APT reconnaissance success rate suppressed in certified Moving-Target-Defence environments.                                                                                   |
| <b>Ecosystem Uptake and Strategic Impact</b> | No EU-headquartered SIEM vendor in the Gartner Leaders quadrant; structural dependency on US platforms; OpenEuroLLM does not yet include a dedicated security stream [32].                                                                                                                                                                                                                                                      | Emergence of certified EU-sovereign alternatives in regulated-sector SOC procurement, with the first export of EU AI security tools to partner jurisdictions.                                                                                                                                                                                                                         | AI Act-derived frameworks for trustworthy autonomous security agents referenced in bilateral digital-partnership agreements; EU AI Office and ENISA jointly contribute to the international standard for security-AI accountability.               |
| <b>Critical Enabler</b>                      | CEN/CENELEC or ETSI TC CYBER adversarial-robustness test suite for security AI, accompanied by AI Act sandbox provisions for security use cases.                                                                                                                                                                                                                                                                                | Operational AI Act conformity-assessment infrastructure for security AI (notified bodies designated under Art 9 plus a published adversarial-robustness test suite) sufficient to certify autonomous security agents for regulated-sector deployment.                                                                                                                                 | Formal adoption of EU security-AI governance by at least one non-EU jurisdiction, treated as the evidence test for the leadership transition.                                                                                                      |

ticle 28 continuous ICT third-party monitoring [12], with the first CTPP designations operationalising the framework [89]; zero-knowledge-proof verification advances from applied research towards demonstrator readiness. The principal Phase 1 risk is regulatory fragmentation: CRA, NIS2, DORA, and the Data Act [27] activate concurrently without a unified interoperability mandate, creating compliance complexity that disproportionately burdens SMEs. The market position is mixed: hyperscalers retain dominant share of the EU cloud market through proprietary API ecosystems [7], supply-chain verification remains vendor-declared under non-EU certification ecosystems, and cross-border AI-model migration requires bespoke bilateral legal review per transfer event throughout the phase.

Phase 2 industrialises federation. Simpl security MIMs are ratified as CEN/CENELEC standards and referenced as

mandatory compliance requirements in the EU Cloud Rulebook, transforming MIM adoption from voluntary best practice into a market-access obligation for providers serving regulated sectors. Automated policy translation is deployed across the majority of active Common European Data Spaces, federated identity attestation is automated, and cross-provider compliance reporting becomes machine-readable through OSCAL [81], [82]. Runtime SBOMs with cryptographic proof-of-execution are deployed in critical infrastructure across multiple Member States, ZKP batch verification operates at scale, and Digital Product Passports become mandatory under the EU Ecodesign Regulation for ICT hardware categories. CAMARA-based cross-border roaming operates across multiple EU operator pairs, and “Data Visa” mechanisms enable dynamic AI-model migration without manual legal review. IPCEI-CIS-aligned NaaS federation

**Table 4.** Cluster 3 transition matrix: Federated Operational Sovereignty.

|                                              | Phase 1 (2026–2030)                                                                                                                                                                                                                                                                                                                                                             | Phase 2 (2030–2035)                                                                                                                                                                                                                                                                                        | Phase 3 (2035–2040)                                                                                                                                                                                                                                                                                                                                                                                                                                          |
|----------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>ODP</b>                                   | Applied R&D → Fragmented Piloting (heterogeneous across sub-capabilities)                                                                                                                                                                                                                                                                                                       | Standardised Integration                                                                                                                                                                                                                                                                                   | Ubiquitous Operation                                                                                                                                                                                                                                                                                                                                                                                                                                         |
| <b>SSL</b>                                   | High Foreign Dependency, with heterogeneity across federation sub-capabilities                                                                                                                                                                                                                                                                                                  | Emerging EU Ecosystem                                                                                                                                                                                                                                                                                      | Full Sovereign Autonomy                                                                                                                                                                                                                                                                                                                                                                                                                                      |
| <b>R&amp;I Maturity</b>                      | Simpl security MIM proof-of-concepts in Common European Data Spaces (Manufacturing, Health, Energy) under manual policy coordination; OASC MIM6 (Security management) progresses from work-item status towards formally published specification; ZKP verification advances from applied research towards demonstrator readiness.                                                | Runtime SBOMs with cryptographic proof-of-execution deployed in critical infrastructure across multiple Member States; ZKP batch verification operates at scale; “Data Visa” mechanisms enable dynamic AI-model migration across European jurisdictions without manual legal review.                       | ZKP-verified supply-chain attestation operates at real-time transaction scale for all critical-infrastructure component verification; Simpl-based security becomes the universal standard for cross-provider Data Space operation.                                                                                                                                                                                                                           |
| <b>Policy, Regulation, and Standards</b>     | CRA Annex I Part II static SBOM disclosure, NIS2 Art 21(2)(h) supply-chain risk assessment, DORA Art 28 continuous ICT third-party monitoring (first CTPP designations, November 2025), and Data Act portability obligations activate concurrently without a unified interoperability mandate; principal risk is regulatory fragmentation that disproportionately burdens SMEs. | Simpl security MIMs ratified as CEN/CENELEC standards and referenced as mandatory compliance requirements in the EU Cloud Rulebook; cross-provider compliance reporting machine-readable through OSCAL; Digital Product Passports mandatory under the EU Ecodesign Regulation for ICT hardware categories. | An EU federated cloud-sovereignty legal framework, grounded in the Commission’s Cloud Sovereignty Framework and future Cloud and AI Development Act [16], resolves US CLOUD Act jurisdictional conflict for federated EU-sovereign workloads; EU Cloud Rulebook mandatory for all new cloud deployments in regulated sectors without exception; EU supply-chain governance standards referenced in ISO/IEC standards and bilateral digital trade agreements. |
| <b>Operational Deployment and Adoption</b>   | Harmonised CAMARA Network APIs deployed in bilateral 5G corridor agreements; cross-border edge-compute roaming not yet operational at scale; supply-chain verification at static disclosure stage; cross-border AI-model migration requires bespoke bilateral legal review per transfer event [90].                                                                             | Automated policy translation across the majority of active Common European Data Spaces; federated identity attestation automated; CAMARA-based cross-border roaming operates across multiple EU operator pairs; IPCEI-CIS-aligned NaaS federation at pilot commercial scale [90].                          | EU NaaS federation operates as a single commercial entity legally equivalent to a unified jurisdiction for data workloads; every new cloud-edge deployment incorporates Simpl MIM compliance, Runtime SBOM verification, and cross-border sovereignty assurance from inception.                                                                                                                                                                              |
| <b>Ecosystem Uptake and Strategic Impact</b> | Hyperscalers retain dominant share of the EU cloud market through proprietary API ecosystems; EU-headquartered providers collectively at a structurally lower share; supply-chain verification vendor-declared under non-EU certification ecosystems [7].                                                                                                                       | First observed decline in foreign hyperscaler share of new regulated-sector cloud procurement; MIM compliance becomes a market-access obligation for providers serving regulated sectors.                                                                                                                  | Foreign hyperscaler new-procurement share below the threshold defined in Section VII; at least two partner jurisdictions formally adopt EU supply-chain governance standards.                                                                                                                                                                                                                                                                                |
| <b>Critical Enabler</b>                      | Simpl-Open MVP consolidation with formal publication of OASC MIM6 (Security management).                                                                                                                                                                                                                                                                                        | Ratification of Simpl MIMs as CEN/CENELEC standards with mandatory reference in the EU Cloud Rulebook, coupled with operational Runtime SBOM attestation in critical infrastructure across multiple Member States.                                                                                         | Enactment of an EU federated cloud-sovereignty legal framework resolving CLOUD Act conflict, with international adoption of EU supply-chain governance standards.                                                                                                                                                                                                                                                                                            |

operates at pilot commercial scale [91]. The strategic shift in this phase is the first observed decline in foreign hyperscaler share of new regulated-sector cloud procurement.

Phase 3 establishes the unified European trust fabric. Simpl-based security becomes the universal standard for cross-provider Data Space operation; ZKP-verified supply-chain attestation operates at real-time transaction scale for all critical-infrastructure component verification; and the EU NaaS federation operates as a single commercial entity legally equivalent to a unified jurisdiction for data workloads. The defining policy instrument is an EU federated cloud-sovereignty legal framework, grounded in the Commission’s Cloud Sovereignty Framework and future Cloud and AI

Development Act [16], that would harmonise sovereignty requirements for cloud and AI computing services across the single market and address US CLOUD Act jurisdictional conflict for federated EU-sovereign workloads [29]; the EU Cloud Rulebook becomes mandatory for all new cloud deployments in regulated sectors without exception, and EU supply-chain governance standards (Runtime SBOM plus ZKP) are referenced in international supply-chain security standards (ISO/IEC) and in bilateral digital trade agreements. Every new cloud-edge deployment incorporates Simpl MIM compliance, Runtime SBOM verification, and cross-border sovereignty assurance from inception. Full fed-



erated sovereign autonomy is conditional on three concurrent outcomes: foreign hyperscaler new-procurement share below the threshold defined in Section VII, enactment of an EU federated cloud-sovereignty legal framework, and at least two partner jurisdictions formally adopting EU supply-chain governance standards.

#### D. PHASE-GATE READINESS OBSERVATIONS

The Phase-1→Phase-2 transition is differently constrained across the three clusters. For Cluster 1, the Critical Enabler (EUCC “High” for an EU-origin open-source RISC-V secure element) is downstream of EU Chips Act fabrication capacity, with adverse signals reported in the European Court of Auditors review [84] and amplified by the cancellation of the Intel Magdeburg plant in July 2025; the gate is therefore at risk on the supply side rather than on the certification side. For Cluster 2, the Critical Enabler (a CEN/CENELEC or ETSI TC CYBER adversarial-robustness test suite for security AI) is the single most consequential standards gap in the cluster and is tightly coupled to documented conformity-assessment body capacity shortfalls [17]; without it, autonomous security agents cannot be certified for regulated-sector deployment regardless of technical readiness. For Cluster 3, the Critical Enabler (Simpl-Open MVP consolidation with formal publication of OASC MIM6) is institutionally on track, with iterative MVP releases since January 2025 [59], but procurement-enforceable only once MIMs are ratified as CEN/CENELEC standards and referenced in the EU Cloud Rulebook in Phase 2. The three Phase 1 gates are therefore constrained by industrial fabrication capacity (Cluster 1), standardisation throughput (Cluster 2), and the sequencing of voluntary specification with binding regulatory reference (Cluster 3). Each requires a different mitigation instrument: dedicated secure-element pilot lines and second-source agreements for Cluster 1; accelerated CEN/CENELEC adversarial-robustness work supported by AI Act sandbox provisions for security use cases for Cluster 2; and early procurement signalling that references Simpl MIMs as preferred pending ratification for Cluster 3.

### VII. STRATEGIC RECOMMENDATIONS

The 2040 destination is a secure, sovereign, interoperable, and resilient European Cloud–Edge–IoT continuum. The seven recommendations below derive directly from the cluster trajectories of Section VI and the phase-gate readiness assessment of Section VI-D. Each recommendation names the action, the cluster or cross-cutting dependency it addresses, the Phase 1 gate it conditions, the quantified Phase 2 and Phase 3 targets it carries, and the principal risk to delivery. Recommendations R1–R3 are cluster-aligned; recommendations R4–R7 address dependencies shared by two or more clusters and require dedicated cross-cluster governance to avoid becoming systemic bottlenecks.

#### A. CLUSTER-ALIGNED RECOMMENDATIONS

a: R1: Sovereign Secure Silicon and Crypto-Agile Infrastructure (Cluster 1).

Combine Horizon Europe Cluster 3 funding with EU Chips Act pilot lines to advance EU-origin open-source RISC-V secure elements (L1.1) towards EUCC “High” assurance, while delivering crypto-agility middleware aligned with the NIS CG PQC roadmap milestones [8] and ETSI TS 103 744 hybrid key-exchange profiles [54] (L5.2). The 2025 EUCC “High” certifications achieved by Infineon, NXP, and STMicroelectronics [30] demonstrate framework readiness; the remaining gap is open-source, EU-controlled fabrication and operationalised PQC migration for edge and IoT. Phase 2 targets: 15–30% EU-designed secure-element share in new critical-infrastructure procurement; hybrid PQC under ETSI TS 103 744 as procurement default for new Industrial IoT and DORA-scoped financial-sector deployments; certified  $\geq 72$  h autonomous Island Mode in cross-border 5G corridors. Phase 3 targets: EUCC “High” coverage of  $\geq 60\%$  of EU critical-infrastructure edge nodes; PQC-native silicon as the EU procurement default;  $\geq 72$  h autonomous Island Mode as a baseline for NIS2-essential and DORA-critical deployments. Principal risk: EU Chips Act fabrication-capacity shortfall (R4).

b: R2: Sovereign Cognitive Cyber Defence with Bounded Autonomy (Cluster 2).

Build on EuroHPC AI Factories [87] and the ECCC Strategic Agenda [85] to deliver EU-sovereign security-focused small language models (CL1.2 and CL2.4), federated SOC and structured CTI operation (CL2.1, CL2.5), bounded autonomous response under ETSI GS ZSM 009 [88] and NIST SP 800-207 [4] (CL2.4 and L4.1), and trustworthy autonomous decision-making aligned with the NIST AI RMF [69], [70] and AI Act Article 9 conformity assessment [11] (CL4.7). Phase 1 target: demonstration of dwell-time reduction below 72 h in at least one certified pilot in a NIS2-essential entity, measured against that entity’s documented pre-pilot baseline. Phase 2 target:  $\leq 72$  h dwell time across certified DORA-compliant, energy, and telecommunications deployments in at least three Member States; EU-sovereign SLM coverage of  $\geq 30\%$  of regulated-sector SOC deployments. Phase 3 targets:  $\leq 24$  h dwell time across NIS2-essential deployments; EU-sovereign AI security tools holding  $\geq 50\%$  of new critical-infrastructure SOC deployments; APT reconnaissance success rate below 5% under continuous Moving Target Defence in certified environments. Principal risk: AI Act conformity-assessment capacity shortfall (R5).

c: R3: Procurement-Enforceable Federation through Simpl, Runtime SBOMs, and Continuous Certification (Cluster 3). Move OASC MIM6 (Security management) from work-item status to formally published specification, then to ratified CEN/CENELEC standard referenced in the EU Cloud Rulebook (CL3.1 and CL3.4), so that interoperability carries procurement-level enforcement. In parallel, advance supply-

chain verification from CRA-mandated static SBOM disclosure [10] to Runtime SBOMs with cryptographic proof-of-execution and zero-knowledge-proof batch verification (CL1.3 and CL4.2), and use EUCC [14] as the foundation while unblocking EUCS [13], plausibly through the 2026 Cybersecurity Act revision (COM(2026) 13) [16], to extend continuous assurance to cloud services. Federate SOC and CTI capabilities across Member States by building on DORA CTPP oversight [89] and cross-border CSIRT cooperation [73]. Phase 2 targets: Simpl MIMs ratified as CEN/CENELEC standards and referenced in the EU Cloud Rulebook; Runtime SBOMs with proof-of-execution deployed in critical infrastructure across at least five Member States; CAMARA-based cross-border roaming across at least five EU operator pairs;  $\geq 30\%$  non-hyperscaler share in new regulated-sector cloud procurement, with foreign hyperscaler share declining for the first time from the approximately 70% baseline [7]. Phase 3 targets: foreign hyperscaler share below 40% in new EU regulated cloud procurement; EU federated cloud-sovereignty legal framework enacted, resolving US CLOUD Act jurisdictional conflict for federated EU-sovereign workloads [29]; at least two partner jurisdictions formally adopting EU supply-chain governance standards. Principal risk: Runtime SBOM attestation maturity (R6).

## B. CROSS-CUTTING RECOMMENDATIONS

a: R4: EU Chips Act Sovereign Fabrication for Security-Critical Components (Clusters 1 and 3).

Five Chips JU pilot lines are operational, with combined funding of approximately €3.7B [83], but no pilot line is currently dedicated to secure-element fabrication; restriction of access to any major non-EU foundry would immediately impair EU critical-infrastructure attestation capability. The European Court of Auditors projects approximately 11.7% global semiconductor share by 2030 against the 20% Chips Act target [84], and the cancellation of the Intel Magdeburg plant in July 2025 removed the largest EU-based advanced-node commitment. Action: establish a dedicated Chips JU pilot line for security-critical components, secure second-source agreements between EU foundries, and implement strategic stockpiling of EUCC-certified secure elements for critical-infrastructure procurement. This recommendation conditions the Phase 1→Phase 2 gate of R1 and the Runtime SBOM hardware-attestation path of R6.

b: R5: AI Act Conformity Assessment Infrastructure for Security AI (Cluster 2).

A material gap between AI Act obligations and conformity-assessment body operational readiness has been documented [17], with AI Act sandbox provisions yet to be operationalised for security use cases. Action: accelerate CEN/CENELEC and ETSI TC CYBER development of an adversarial-robustness test suite for security AI, designate national notified bodies for security-AI conformity assessment under AI Act Article 9, and operationalise AI Act

sandbox provisions for security use cases. This recommendation conditions the Phase 1→Phase 2 gate of R2; without it, autonomous security agents cannot be certified for regulated-sector deployment regardless of technical readiness.

c: R6: Runtime SBOM Attestation Beyond CRA Static Disclosure (Clusters 1 and 3).

Static SBOMs, mandated by CRA Annex I Part II from December 2027 [10], are insufficient for dynamic continuum operations in which containerised microservices and AI components migrate across providers and jurisdictions. Action: progress from static SBOM disclosure to cryptographic proof-of-execution mechanisms, with zero-knowledge-proof batch verification operates at scale by Phase 3, anchored in the hardware roots of trust matured under R1 and aligned with the IPCEI-CIS ICRA Reference Architecture [91]. This recommendation is a shared Phase 2→Phase 3 transition condition for R1 and R3, and the principal mechanism by which Cluster 1 hardware attestation is operationalised for Cluster 3 federation.

d: R7: SME Inclusion across Sovereign Hardware, Cognitive Defence, and Federation (all clusters).

Cost barriers exclude SMEs from sovereign hardware alternatives, AI-powered SOC tooling, and Simpl MIM compliance, threatening to make NIS2, CRA, and AI Act compliance a structural barrier to innovation. Action: implement tiered certification pathways under EUCC and EUCS, subsidise shared cyber-range and digital-twin access (CL3.5), provide reusable compliance tooling (OSCAL-based control catalogues [81], [82]), and target Digital Europe Programme co-funding at shared-infrastructure pathways for SME access to sovereign secure elements, EU-sovereign security SLMs, and Simpl MIM-compliant federation services. This recommendation conditions the ecosystem-uptake dimension of all three cluster trajectories: without SME participation, sovereign capabilities risk being deployed only by large industrial actors and public bodies, leaving the broader European industrial base on non-sovereign alternatives.

## VIII. DISCUSSION, LIMITATIONS, AND FUTURE WORK

The roadmap is an evolving strategic artefact rather than a deterministic forecast. Its value lies in providing a shared analytical structure through which stakeholders can align research, policy, deployment, and investment decisions around a common set of objectives, and in making cross-cluster dependencies and phase-gate conditions explicit enough to support portfolio-level decision-making under conditions of ecosystem uncertainty. Validation is conducted through expert workshops within the NexusForum and EuCloud-EdgeIoT Working Group on Cybersecurity, consultation with national cybersecurity agencies and industry, and policy dialogue with European Commission services, addressing completeness and relevance of the architectural decomposition, clarity and non-overlap of capability and priority formulations, coherence of priority clustering, credibility of tran-

sition conditions, and reasonableness of Sovereignty Level assessments.

Three limitations bound the analysis. First, the Cloud–Edge–IoT landscape is shaped by rapidly evolving technologies, regulatory implementation, market dynamics, and geopolitical conditions, all of which may alter cluster trajectories; the roadmap should be read as a strategic synthesis of the state of the art and expert judgement rather than a precise prediction. Second, the Sovereignty Level construct is used as an interpretive lens rather than a quantitative score, and its operational comparability across clusters depends on identifying measurable proxies for strategic dependency. Third, quantified targets in Section VII are anchored in 2025–2026 evidence and require revision as Phase 1 enforcement, certification, and procurement signals accumulate.

Future work follows three directions. First, the architectural analysis should be instantiated in sector-specific use cases, including energy, healthcare, finance, and public procurement, to examine how cybersecurity priorities are affected by sector-specific regulation, procurement rules, supply-chain constraints, and market structure, capturing dependencies that fall outside a cybersecurity-centred perspective. Second, the Sovereignty Level construct should be further operationalised through measurable proxies such as provider concentration, certification availability, fabrication capacity, and cross-border deployability, and validated through cross-cluster comparison. Third, the roadmap should be maintained as an iterative and dynamic instrument, periodically revised as new empirical evidence, stakeholder feedback, regulatory milestones, and technological developments become available, with explicit revision triggered by Phase 1→Phase 2 gate-readiness reassessments.

## IX. CONCLUSION

The Cloud–Edge–IoT continuum requires Europe to move beyond reactive and fragmented cybersecurity models towards autonomous, federated, and certified Security-by-Design. The three-stage methodology presented here decomposes the continuum into six Security Layers and four Transversal Security Cross-Layers populated with their constituent capabilities (Stage 1), consolidates those capabilities into three executive priority clusters analysed through a seven-dimension template (Stage 2), and projects each cluster onto a 2026–2040 transition matrix in which a single Critical Enabler per phase column functions as the gate condition for progression (Stage 3). The seven recommendations of Section VII, comprising three cluster-aligned and four cross-cutting recommendations, trace each quantified target back to the phase gate it depends on, making the ambition–implementation gap operationally actionable rather than rhetorical.

The 2025–2026 evidence base shows concrete milestones alongside persistent gaps. EUCC “High” certifications across the three major EU secure-element designers [30], a binding NIS CG PQC migration framework [8], DORA operationalisation through the first CTPP designations [89], and the itera-

tive Simpl-Open MVP [59] establish Phase 1 readiness in domains where standardisation and certification are the binding constraints. At the same time, NIS2 transposition delays [15], EUCS deadlock [16], contested Chips Act semiconductor targets [84], AI Act conformity-assessment capacity shortfalls [17], and continued hyperscaler concentration in the EU cloud market [7] mark the gates where the Phase 1→Phase 2 transition is at risk. The Phase 1 Critical Enablers identified for each cluster, namely EUCC “High” for an EU-origin RISC-V secure element, a CEN/CENELEC or ETSI TC CYBER adversarial-robustness test suite for security AI, and ratification of OASC MIM6 with EU Cloud Rulebook reference, are constrained in different ways and require tailored mitigation instruments.

Future work should extend this analysis in three directions: empirical validation and operationalisation of the Sovereignty Level construct through measurable proxies for provider concentration, certification availability, fabrication capacity, and cross-border deployability; sector-specific instantiation of the cluster trajectories in energy, healthcare, financial services, and public procurement, where sector-specific regulation and supply-chain structure modify the generic phase logic; and integration of the recommendations with the next EU Framework Programme funding landscape, with periodic gate-readiness reassessment as Phase 1 enforcement, certification, and procurement signals accumulate.

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